Introduction to MS Project

Lab 1 – Project Programming (WBS, PLE, CPM, Resources, Cost)

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1 Introduction

In this tutorial, you will learn the main (yet most used) features of MS Project:

- How to setup a new project,
- Develop a WBS
- Create the Project Logic (PLE),
- Set durations to Tasks
- Conduct CPM Network Analysis (time).
- Creating and Assigning Resources
- Resolving Over-allocations and Levelling
- Entering Cost Information
- Visualisation and Printing of Programme, and Production of Report

It is important to keep in mind that MS Project is one of the several available tools for project planning. MS Project is not one of the most advanced programming software packages, but it provides the most commonly used features, and is thus simpler for your learning process. Primavera P6 (EPPM) is an example of more advanced scheduling software package.

To conduct this Lab, please read each main section, looking at the different features in MS Project, but do nothing until you reach the "Apply this" boxed sections. Only do what is suggested in those "Apply this" sections. Do not jump sections or try to do more than what is asked in the given Section; simply follow the document.

2 Setting up a Project

Once you have started MS Project, go to ***File***→Options (see Figure 1). The ***Project Options*** window that opens has multiple sections. Go to the section *Schedule*. This contains multiple options, two of which mainly interest us:

1. *Default Task Type*: this is an important issue. I actually suggest that you leave it as *Fixed Units* for now. But, we will change this latter on. The difference between *Fixed Units*, *Fixed Work* and *Fixed Duration* is discussed in **Section 8.2.1**.
2. *New Task Created*: you want to set this option to *Auto Scheduled*. In MS Project, it is possible to define whether a task will be automatically managed by the software or managed manually. When managed manually, changes to a predecessor will not impact the activity.

You can now close the ***Options*** window.

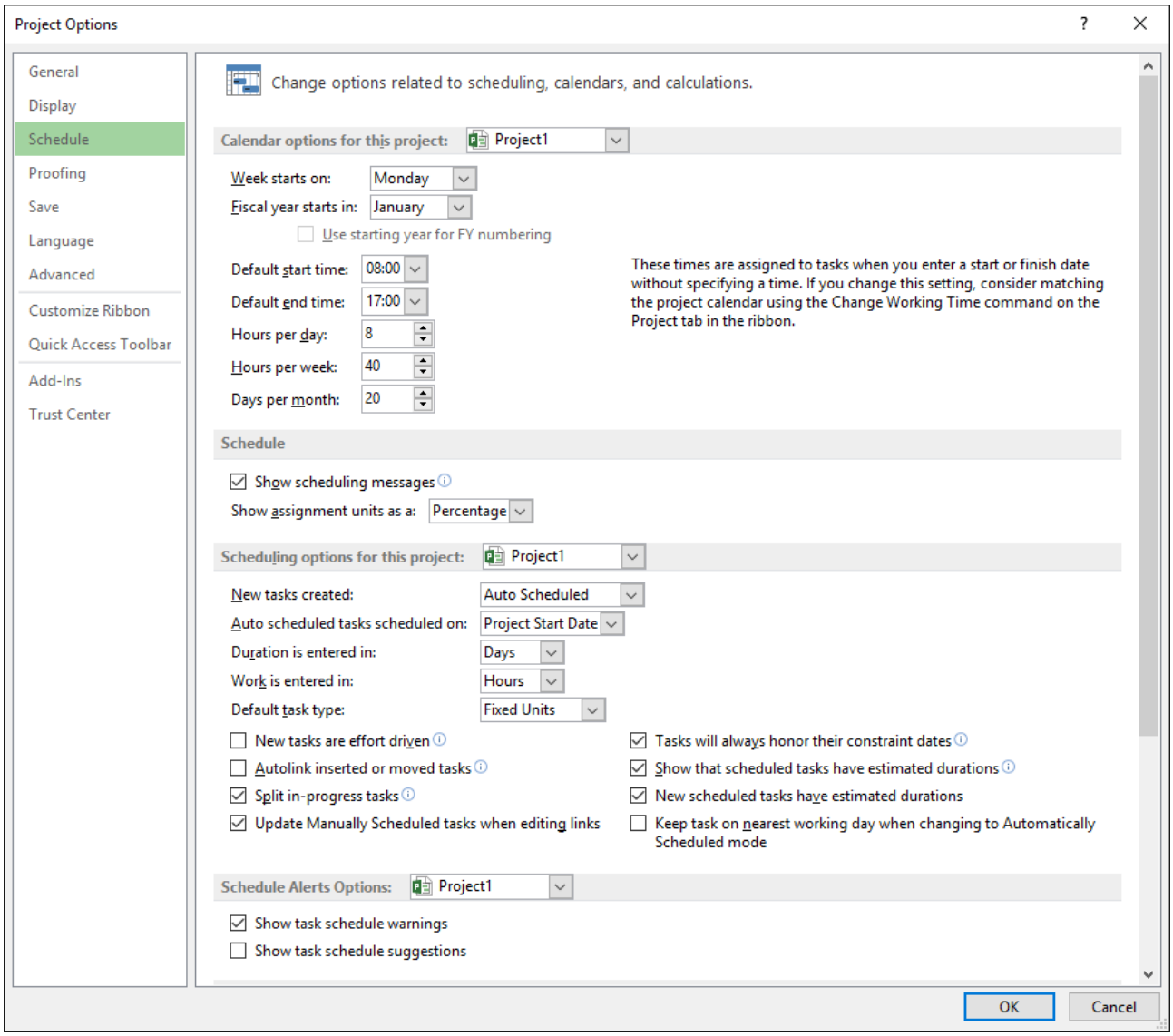


Figure 1: Project Options/Schedule window.

Then, the first step in the development of your project is to set its overall start date. For this open the ***Project Information*** window located in the ***Project*** *ribbon*. In this window (see Figure 2), you can define the programme *Start date*, but also other parameters such as the type of *Calendar* to be used (see **Section 2.1** below), the default level of priority (see **Section 9**) and the current date (used during project Control).

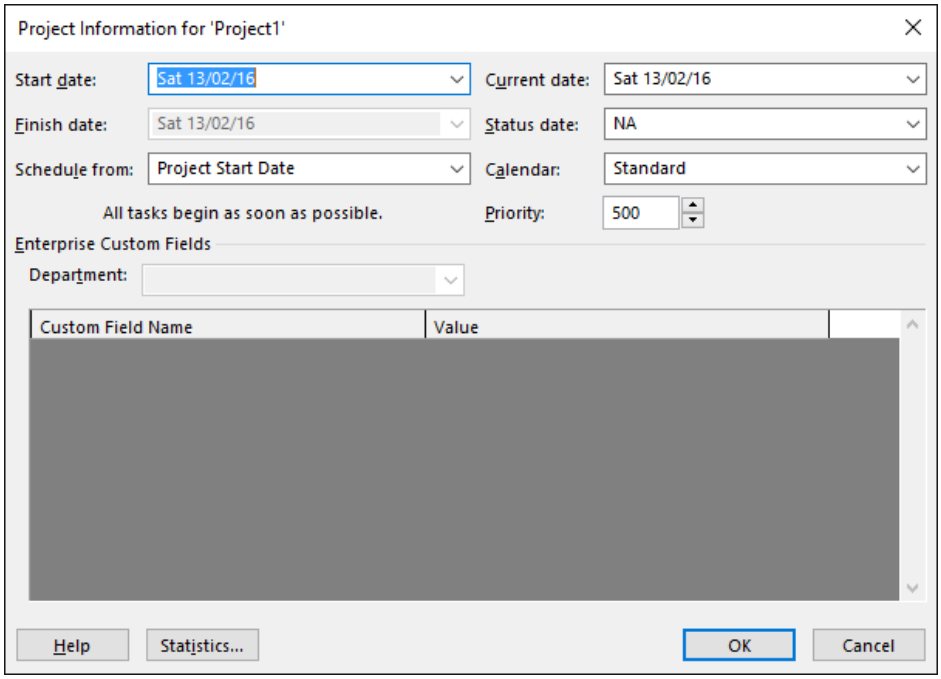


Figure 2: Project Information window

2.1 Defining the project calendar

When setting up a project, baseline working times and calendar dates have to be clearly defined: holidays cannot be missed nor a day off. To define the project calendar, go to ***Project***→*Change Working Time*. In the ***Change Working Time*** window (see Figure 3), you find the *Standard* working calendar but also two other calendars (*Night shift* and *24 hours*) and a calendar for each resource. The Standard calendar is a 5-day work week from 8am to noon and 1pm to 5pm (8 hour per day) with Saturdays and Sundays off (40 hours per week).

You can add exceptions, e.g. to record days when the working schedule may be different or to block off holidays and other days off. This is done by adding *Exceptions* to the calendar (see Figure 3a). You can also create your own specific calendar if you wish, as described below.

If you need to change the work daily and weekly work pattern of the calendar selected in the ***Change Working Time*** window, go to the *Work Weeks* tab in the Change Working Time window (see Figure 3b). You can then click on the *Details…* button. Here you can select days and then set whether they non-working or working days, and change their working hours.

If you need to create an entirely new calendar (which is generally preferable instead of making significant changes to default ones), click on *Create New Calendar*. This opens a small window to give the new calendar a name and select whether it should be initiated as a copy of an existing calendar. Once the calendar is

created, you can set it. In particular, you will that, in contrast to the Standard calendar, the *Details…* button is available.

As will be discussed later in this document, MS Project enables the user to associate calendars to different things:

- *Project calendar*: this is overall baseline calendar applied to all tasks and resources by default.
- *Task calendar*: it is possible to set a calendar specifically to any task (see **Section 5.1**).
- *Resource calendar*: similarly, it is possible to also set a calendar specific to a given type of resource (see **Section 8.1)**.

The question then arise which calendar is select to schedule any part of the project. The answer is that those calendars are considered with a fixed priority:

1. *Task Calendar (Highest Priority)*: If you assign a specific calendar to an individual task, MS Project uses it to determine working and non-working times, completely overriding the project calendar. It can also be set to ignore resource calendars.
2. *Resource Calendar (Medium Priority)*: If no task calendar is assigned, Microsoft Project schedules the task based on the calendars of the resources assigned to it. If multiple resources with different calendars are assigned, it uses only the common working times shared by all of them
3. *Project Calendar (Lowest Priority)*: This serves as the default baseline. It dictates the schedule before any resources or task-specific calendars are applied, and acts as the fallback when no other specific calendar constraints exist

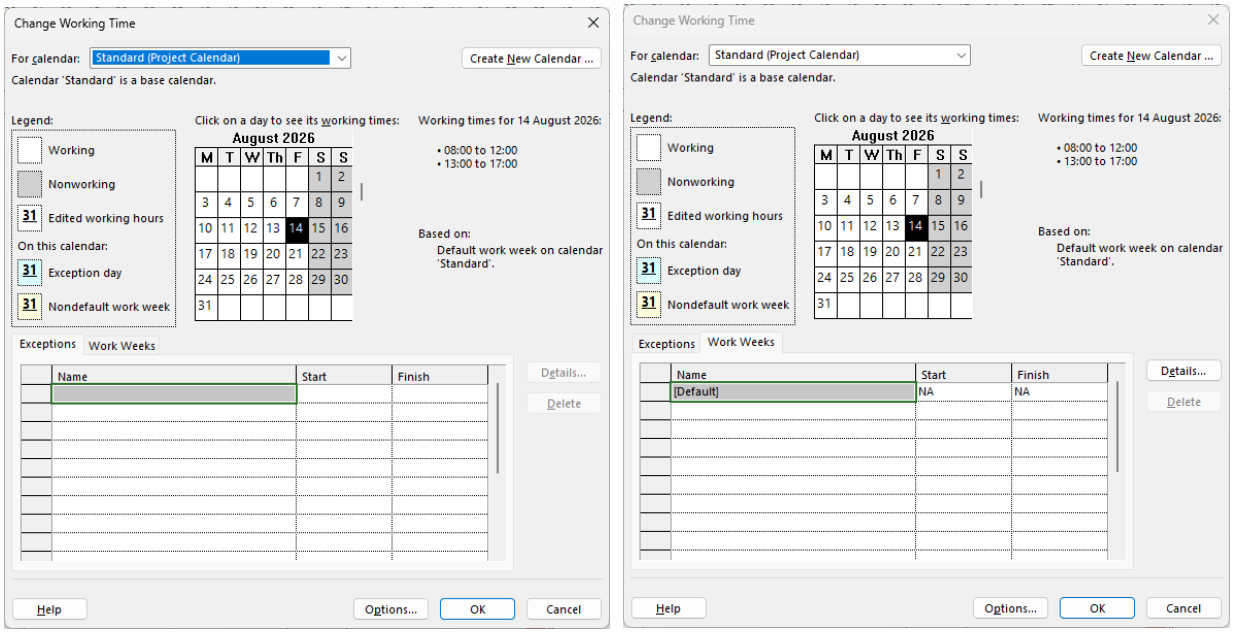


Figure 3: Change Working Time window.

Apply this:

Using the project information in the attached Support Project sheet, do the following:

1. Open the *File→Options→Schedule* window and set the *Default Task Type* to *Fixed Units*, and set *New Task Created* to *Auto Scheduled*, as shown in Figure 1. The Calendar should not be changed, i.e. it should remain as a standard working week calendar.
2. Open the *Project→Project Information* window and set both the *Start Date* and *Current Date* to the project start date defined on your project. Keep the *Standard* calendar the scheduling from the *Project Start Date*.

3 Creating a WBS[1]

3.1 Inserting Tasks

In the Gantt Chart view that you are seeing by default (see Figure 4), you can see a table on the left side and a Gantt Chart calendar on the right. Each line in the table corresponds to a Work Package/Task/Activity and has an associated bar in the Gantt Chart. To create a Task, select the first line in the table, and click on *Task Information* in the *Task* ribbon – alternatively simply double-click on the line. This opens the *Task Information* window for this particular task. As shown in Figure 5, the *Task Information* window has 6 tabs that will be explained in detail in this and the next lab. For now, you only need to enter the task name in the textbox *Name* (note that the textboxes *Name* and *Duration* remains visible in all tabs).

Note that, as an alternative to the methods above, you can also directly type the task name in the cell that falls in the column *Task Name*.

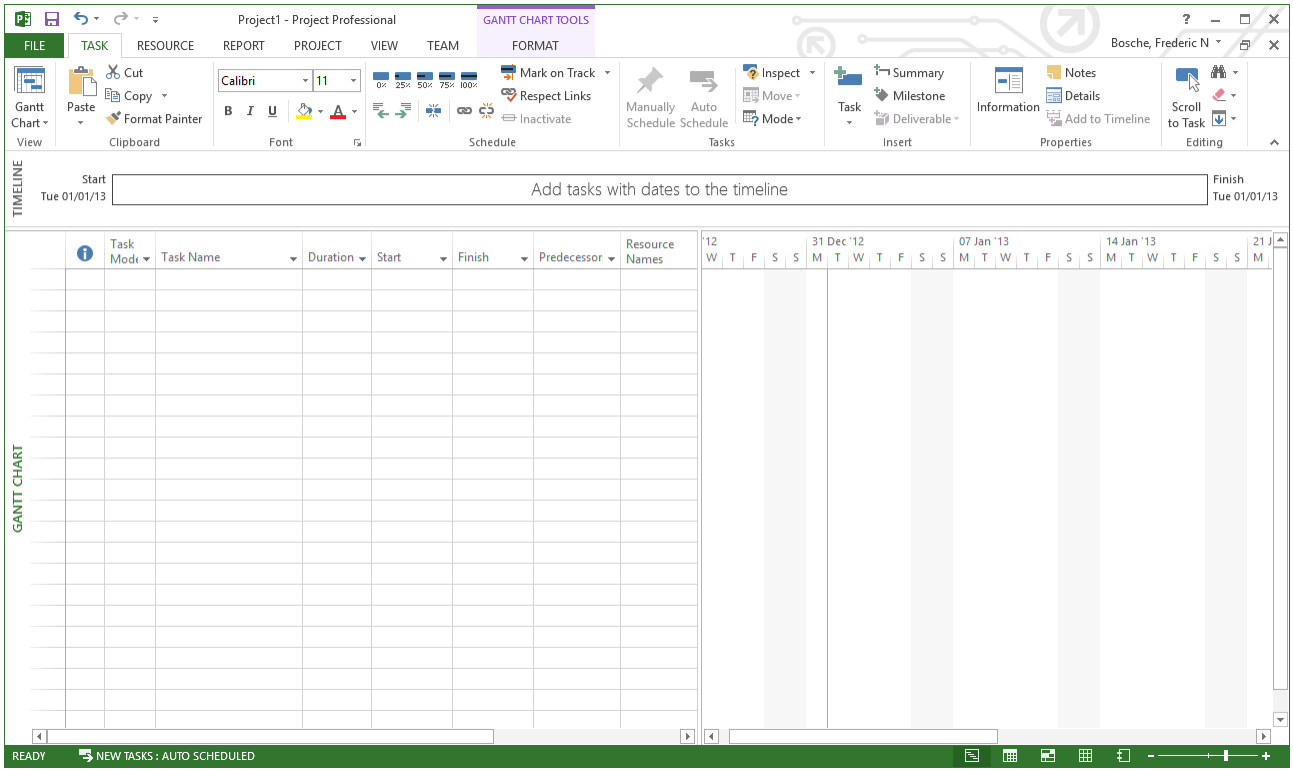


Figure 4: Main window - Gantt Chart view.

1 http://www.project-tutorial.com/how-to-create-and-cod-a-wbs/

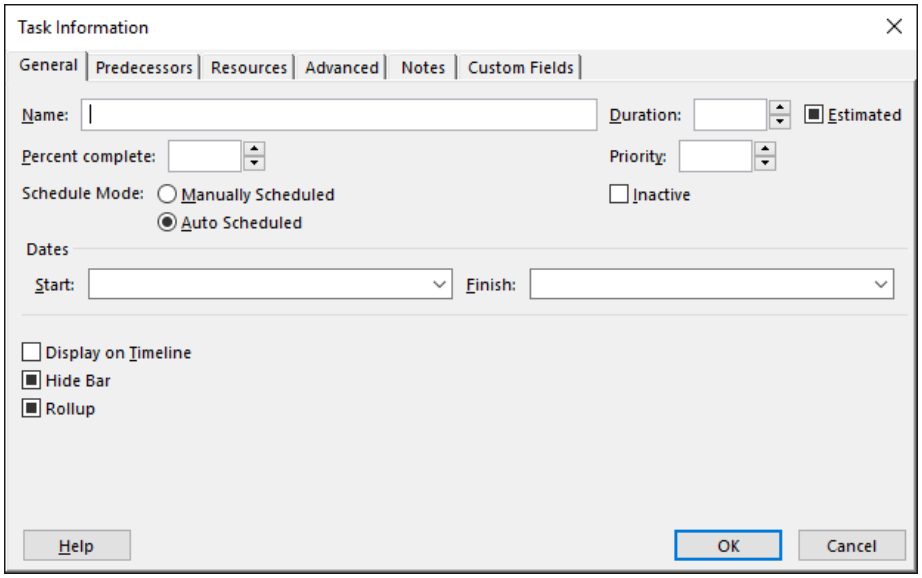


Figure 5: Task Information window - Tab "General".

3.2 Outlining the WBS structure

To correctly enter all items of the WBS, it is important to reproduce its hierarchy faithfully (i.e. Level 0, level 1, etc.). For example, the first entered *task* could be the project name (level 0) under which all the other Work Packages and tasks (level 1, etc.) should appear. Alternatively, it could be the first Level 1 Work Package.

To create this hierarchy, enter the higher-level WP in one line (e.g. Level 1) and the WPs or Tasks under it (e.g. Level 2) in the following lines. Then, select all those lower-level WPs/Tasks (i.e. level 2) and click the *indent* button ( ) in the *Schedule* section of the ***Task*** ribbon. This effectively brings all these WPs/Tasks to level 2. You can similarly create the level 3, 4, etc., and consequently construct the whole WBS. Note that, tasks can also be brought up a level by clicking on the *outdent* button ( ).

3.2.1 WBS Codes

There are two ways to show a set of WBS codes. You may simply wish to use the automated outline numbering (i.e. automated WBS codes) of MS Project that uses the format “1.1.1.1”.

You can show the Outline Numbers in two different ways:

- By adding the *Outline Number* columns to the table. For this, right-click the heading of the column that you want to appear to the right of the *Outline Number* column, select *Insert Column* in the menu and finally select *Outline Number* in the list of possible columns. This column will be automatically filled when you add new activities; or
- By having the *Outline Numbers* be automatically added to the task names. For this, select any cell in the table, then go to the ***Gantt Chart Format*** ribbon, and there check the *Outline Number* checkbox.

However, this standard formatting of Outline Numbers may not reflect the WBS coding used for this project. Don’t worry; it is in fact possible to define and use your own WBS. To do this, go the ***Project*** ribbon, and select *Define Code* in the *WBS* menu. The window shown in Figure 6 then appears that enables you to define

the format of the WBS using a project code, and various code formats for the different levels of the WBS (level 1, level 2, level 3, etc.). The window in Figure 6 shows an example of WBS.

Once the WBS is created, add the *WBS* column to the Gantt Chart table. For this, right-click the heading of the column that you want to appear to the right of the *WBS* column, select *Insert Column* in the menu and finally select *WBS* in the list of possible columns. This column will be automatically filled when you add new activities.

Note: sometimes, MS Project fails to do those updates correctly. If you experience such issue, simply go back to the *WBS* menu again and select *Renumber*…

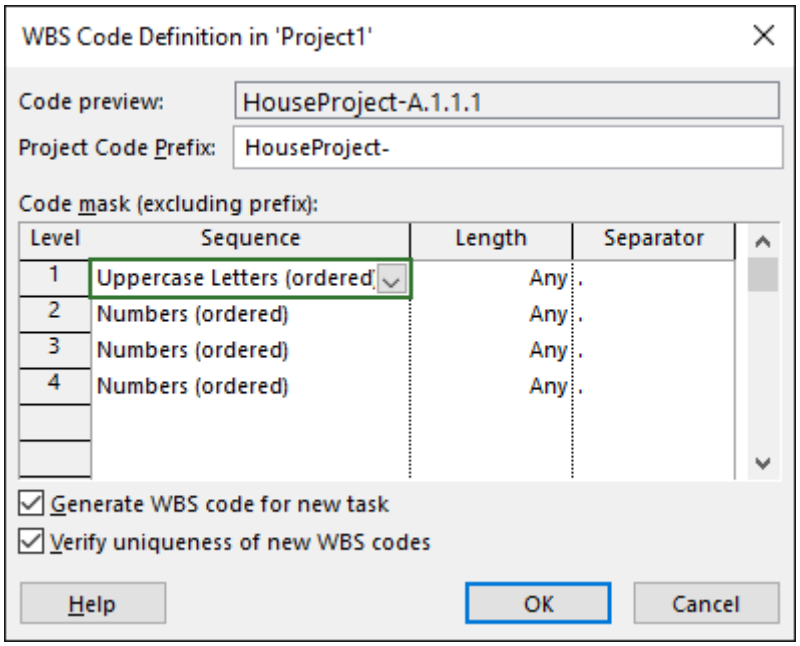


Figure 6: WBS Code Definition window

3.2.2 Project Summary Task

It is often useful to have a Task line at the top of the programme that encapsulates the overall project (e.g. to get overall start and end dates, overall costs, etc.). This could be done by manually entering the task in Line 1 and indenting all the project tasks underneath it. This is actually not needed as MS Project actually keeps in memory this overall information and can directly display it. For this, with the Gantt View open, go to the ***Gantt Chart Format*** ribbon and tick the *Project Summary Task* box located in the *Show/Hide* section. This adds a Project Summary Task in Line 0.

Apply this:

Using the project information in the attached Support Project sheet, do the following:

1. Create a WBS like the one suggested in Figure 6, with the Project Code Prefix "House-", level 1 codes as Uppercase Letters, and all other levels as Numbers (see the project information in the attached Support Project sheet to ensure you create enough levels).

2. Add the WBS column on the left of the Task Name column in the Gantt Chart table.
3. Enter the whole WBS with all Work Packages and Activities for the Support Project. Note that, since we used "House-" as the Project Code Prefix, there is not really a need to add it to the WBS. But, this is up to you.
4. Do not enter manually a Task for the overall project. Instead, just show the Project Summary Task internal to MS Project.

3.3 Creating milestones

In addition to the WBS, one may be interested in inserting Milestones, which are key points in the project delivery. To create a milestone, simply create it as an activity and assign to it a duration of 0 day. Alternatively, create the activity, open its ***Task Information*** window, go to the *Advanced* tab and check the checkbox *Mark task as milestone*. In the Gantt Chart view, the milestone is displayed by default with a black diamond. Note that you can manipulate milestones like any other activity. In particular, precedence relationships can be created with them.

4 Creating the Project Logic (PLE)

Assigning task duration and creating precedence relationships can all be done in the ***Task Information*** window. In that window, the tabs *General*, *Predecessors* and *Advanced* are of particular interest.

There are three main alternative ways to create the precedence relationships making up the project logic.

- Select the task for which you wish to provide the list of predecessors and open its ***Task Information*** window (as shown earlier in Figure 5). In this window, select the tab *Predecessors* (see Figure 7). Each line in the table is for a possible predecessor. To enter one predecessor, select the cell in the column *Task Name* in the first empty line. A dropdown menu appears that lists all the tasks in the project. You can select a predecessor. If the task has more predecessors, select another one in the line underneath.

  Note that, when you select the predecessor, MS Project establishes by default a *Finish-to-Start (FS)* relationship with zero lag. These can be altered as required. Once you confirm your changes (*i.e.* by clicking *OK*), you can see in the Gantt Chart that the bar of the activity has now moved and it is linked to its predecessor with a precedence relationship arrow.

- For the task for which you wish to provide the list of predecessors, select its cell under the Predecessor column in the table. There, type the line number (in the Gantt Chart table) of the preceding activity. If you wish to enter multiple predecessors, simply separate the numbers with commas. Note again that by default the relationships are of type Finish-to-Start with 0 lag.

- Create the precedence relationship directly in the Gantt Chart view. For this, select the preceding task, hold the selection (i.e. don't release the left button of your mouse), and drag the task onto one of the task of interest. After releasing the mouse's button, a precedence relationship arrow is created. By default the relationship is of type Finish-to-Start with 0 lag. This can be changed by double-clicking on the relationship arrow in the Gantt Chart view, or by opening the ***Task Information*** window of the successor activity.

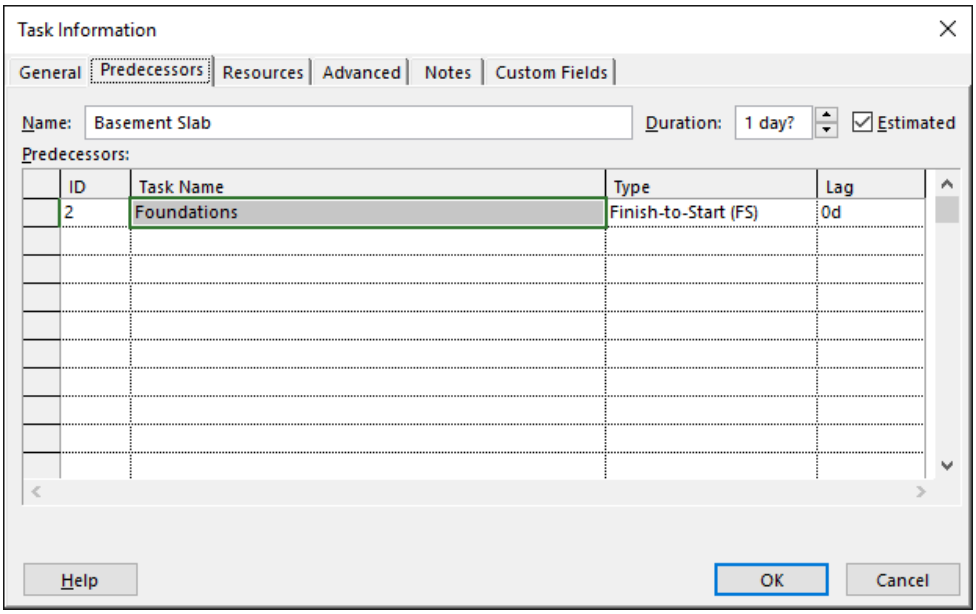


Figure 7: Task Information window - Tab "Predecessors".

5 Assigning Durations to Tasks

Assigning durations to a task can be done in the *Duration* column in the Gantt Chart table, or in the **Task Information** window of each task (see Figure 5 or Figure 7). Default durations are in days, but different units can be used. These include (in brackets are the shortkeys for the units):

- minutes [m]
- hours [h]
- days [d]
- weeks [w]
- months [mo]
- years [y]

Remember that durations only need to be given for the activities in the WBS, not the upper-level Work Packages in the WBS. The durations for the upper-level Work Packages of the WBS are inferred automatically.

5.1 Set a specific calendar to a task

All tasks use by default the Project calendar. But it is possible to overwrite this with a specific calendar for a given task. To do this, create a new specific calendar as explained in **Section 2.1**. Then, open the ***Task Information*** for the task you wish to change the work calendar of. Go to the *Advanced* tab and in the Calendar section, select the specific calendar you just created. Close the *Task Information* window. The task will now follow that specific calendar and not the overall project one.

6 CPM Network Analysis

After having defined the WBS, conducted the Project Logic Evaluation (PLE), and assigned durations to tasks, the CPM programme should appear complete in the Gantt Chart view (see Figure 8). If the Critical Path is not already shown, you can make it visible by selecting any cell in the Gantt Chart table, and then ticking the checkbox *Critical Tasks* under the ***Gantt Char Format*** ribbon.

If you wish to see detailed information about the activities (*e.g.* float/slack, ES, EF, LS, LF), go to a ***View*** ribbon, and select the option *Schedule* in the *Tables* menu. This table gives: ES, EF, LS, LF, FF and TF.

Apply this:

Using the project information in the attached Support Project sheet, do the following:

1. Create the Project Logic for the project using either of the methods above (try all to see which one you prefer)
2. Add the durations to all project activities (see how the durations, start and finish of the upper-level Work Packages in the WBS are calculated automatically).
3. Show the Critical Path of your project. Your programme should look like the one in Figure 8.

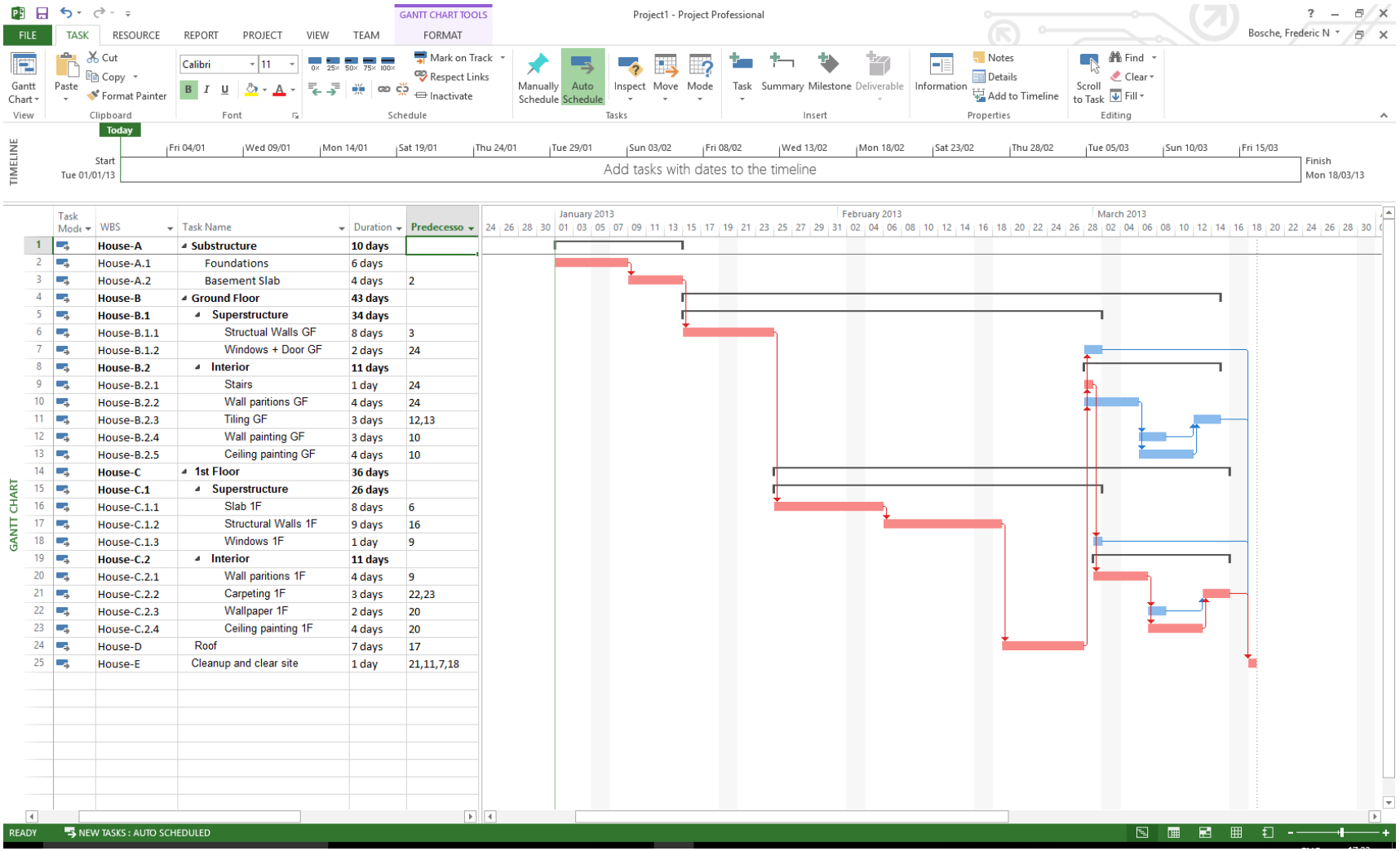


Figure 8: CPM result after creating a WBS, a PLE and defining task durations.

7 Project Views

Before we continue our programming work, it is important to realise that MS Project provides numerous different ways of viewing a project. By default (i.e. when you start MS Project), MS Project shows the ***Gantt Chart*** view, and all we have done so far was in that view. But, many other views exist that can be useful for certain tasks during the planning or monitoring and control of a project. To select a different view, go to the ***Tasks*** ribbon, and open the dropdown menu under *Gantt Chart* (on the left of the ribbon). The views include:

- *Calendar*: you will rarely use this one.
- *Gantt Chart*: most commonly used.
- *Network Diagram*: used to perform and/or review the Project Logic – this is similar to the Network Precedence Diagram (NPD) we have seen in class.
- *Resource Graph/Sheet/Usage/Form*: four different views of resource usage per resource – see Section 8 and **Section 8**.
- *Task Usage/Form/Sheet*: three views of resource usage per task – see **Section 8**.
- *Team Planner*.
- *Timeline*: to create a highly summarized view of a project.
- *Additional views...*: many other views are available.
- *Tracking Gantt*: used during the control stage – see **Lab 2**.

Alternatively, you can go to the ***View*** ribbon. There, all the views are grouped into two main categories related to their purpose: *Task Views* and *Resource Views*.

- *Task* views are more appropriate in the first scheduling stage (entering and linking tasks).
- *Resource* views are better for the following programmes stages, i.e. once resources have been entered (e.g. we will use these views in **Section 8** and **Section 10**).

As introduced earlier, the table in the left side of the Gantt Chart view can be changed into pre-set tables for specific purposes. For example, the default table for the *Gantt Chart* view is *Entry* as it is most appropriate for the entering programme tasks. But, it can be changed by going to the ***View*** ribbon, and open the *Tables* dropdown menu. The following tables can, for example, be selected:

- *Schedule* table: to show the ES, EF, TF, etc.;
- *Cost* table: when entering and tracking cost data; or
- *Tracking* table: when tracking schedule progress.

We will use several of these tables later in this lab and in **Lab 2**.

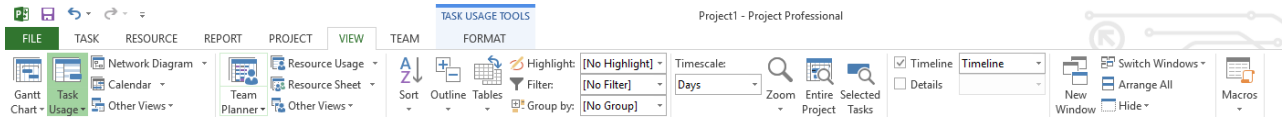


Figure 9: The ribbon View.

Apply this:

There is nothing to be done specifically with the Support Project. However, I recommend you visit the different views and tables discussed above.

8 Creating and Assigning Resources

8.1 Creating Resources

To define all your resources, change from the *Gantt Chart* view to the *Resource Sheet* view (see Figure 10), for example by selecting this view in the ***View*** ribbon. This view is essentially a spreadsheet, in which each line corresponds to one resource. A resource can be defined as representing a specific person (e.g. John Smith), or a type of trade worker without specific identification (e.g. structural labourer), or a piece of equipment (e.g. crane).

For each resource, the following information (columns) can be provided:

- *Type*: This is the type of resource. A "Work" resource is anyone or anything that works on a task in your project. This can be a carpenter, a plumber or a piece of equipment like a mobile crane. "Material" resources are used for things that are consumed by a task, not the things that perform the task.
- *Material Label*: this gives the label/unit for the resource (e.g. yards, tons, or boxes).
- *Initials*: self-understandable.
- *Group*: Resources that share characteristics and are categorized by a group name. For example, you can categorize resources by job function and use a group name such as plumbers. Or, you can categorize resources by employment status.
- *Max. Units*: In MS Project terms, *Units* refer to "Units of resource". *Max. Units* is the percentage of the resource's full working day that the resource is available to do project work. *Max. Units* is used to determine if a resource is over-allocated or under-allocated. MS Project will see a resource as overallocated if, for any day, the cumulative required effort for a resource (i.e. total *Units*) is greater than its availability (*Max. Units*). Here's an example: a carpenter is available only 50% part-time on a standard calendar (Max Units is 50%). Then, if this carpenter is assigned to two concurrent tasks requiring it to work both on a 50% basis, then the carpenter is effectively allocated 100% and the system will identify the worker as being over-allocated. But, if the tasks are not concurrent, then the carpenter is working to its full capacity/availability.

  Since a resource can be an individual, a crew, or a type of individual/crew (e.g. "type of trade worker without specific identification" like "Carpenters"), it is up to you to define what "*Max. Units* =100%" means. For example, it could mean "3 carpenters" (and therefore 33% would mean 1 carpenter). Or you could assume that each carpenter is equivalent to 100% and therefore you would set "*Max. Units* =300%".
- *Std. Rate*: See **Section 10**.
- *Ovt. Rate*: See **Section 10**.
- *Cost/Use*: See **Section 10**.
- *Accrue At*: See **Section 10**.
- *Base Calendar:* You can select the calendar defining the work pattern of each resource individually. **Section 2.1** discussed how different calendars can be created and/or altered.

- *Code*: If you have a special code for this resource.

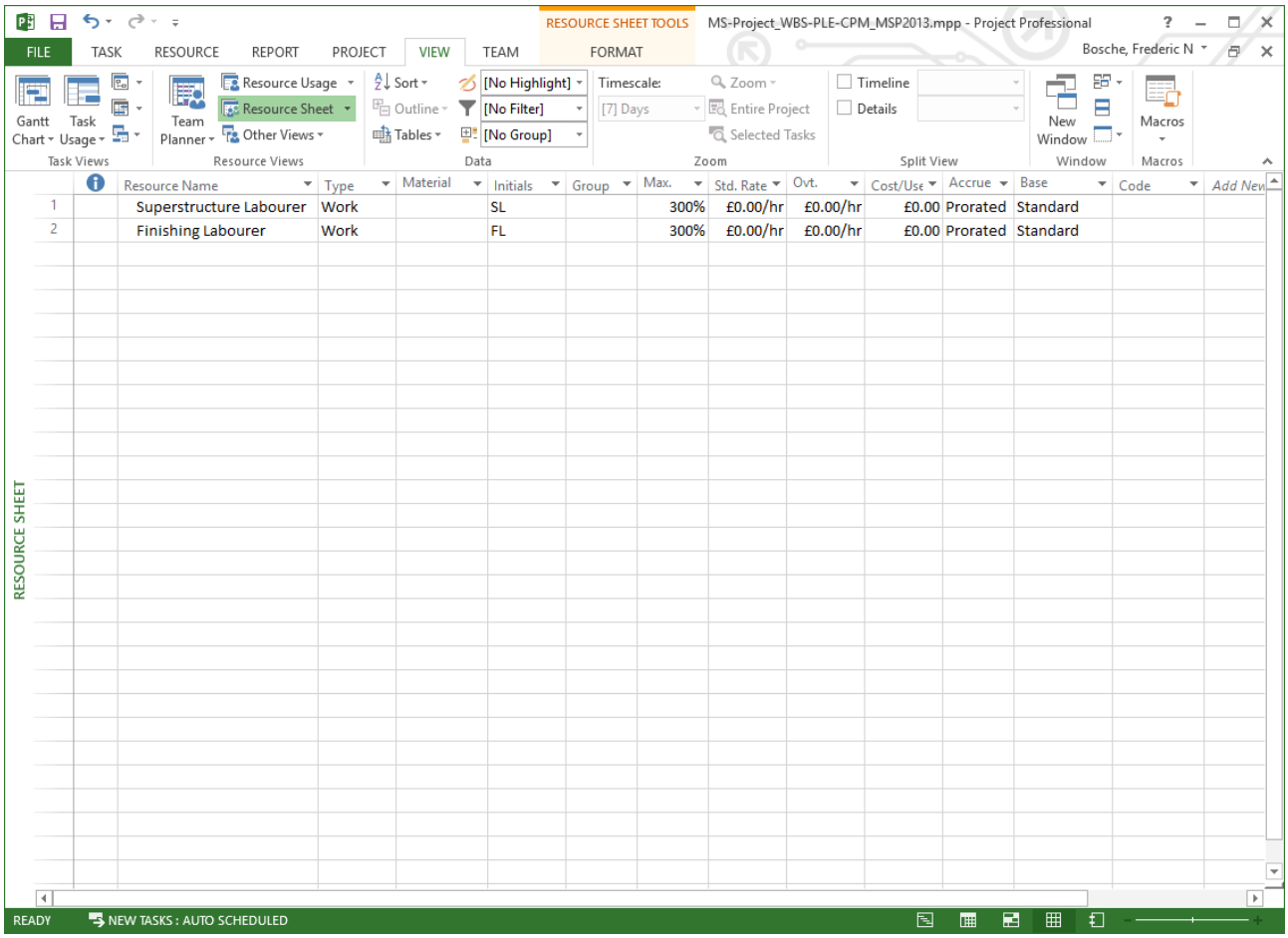


Figure 10: Resource Sheet view.

Apply this:

Using the project information in the Support Project sheet (from Lab 1), do the following:

3. Open the resource sheet view and create the resources available for your project along with their maximum availabilities.

8.2 Assigning Resources

8.2.1 Relationship between Units, Duration and Work

Before explaining how to assign resources to each activity, it is very important to discuss a matter that is the source of significant confusion among MS Project users (me included, at times). So, take your time reading this section carefully.

First, it is important to understand that there is a relationship between the *Maximum Daily Workload* available for a given resource *(hours/day/unit)*, the maximum number of *Units (unit)* of that resource assigned to a given task, the amount of resource *Work (hours)* necessary to perform that task, and the resulting task *Duration (days)*. The formula is:

$$Duration = \frac{Work}{Maximum\ Daily\ Workload \times Units} \qquad (1)$$

The *Maximum Daily Workload* is defined by the calendar associated to the given resource, as discussed in the Sections 2.1 and 8.1. As discussed in Section 2.1, the *Standard* calendar is used by default with work happening Monday to Friday for 8 hours each day. That calendar can however be changed to better represent the working pattern of that resource.

To particularly see the behaviour of *Duration* and *Work* when you schedule your project, you can add the column *Work* next to the column *Duration* in your programme *Entry* table (e.g. in *Gantt Chart* view), as shown in Figure 11.

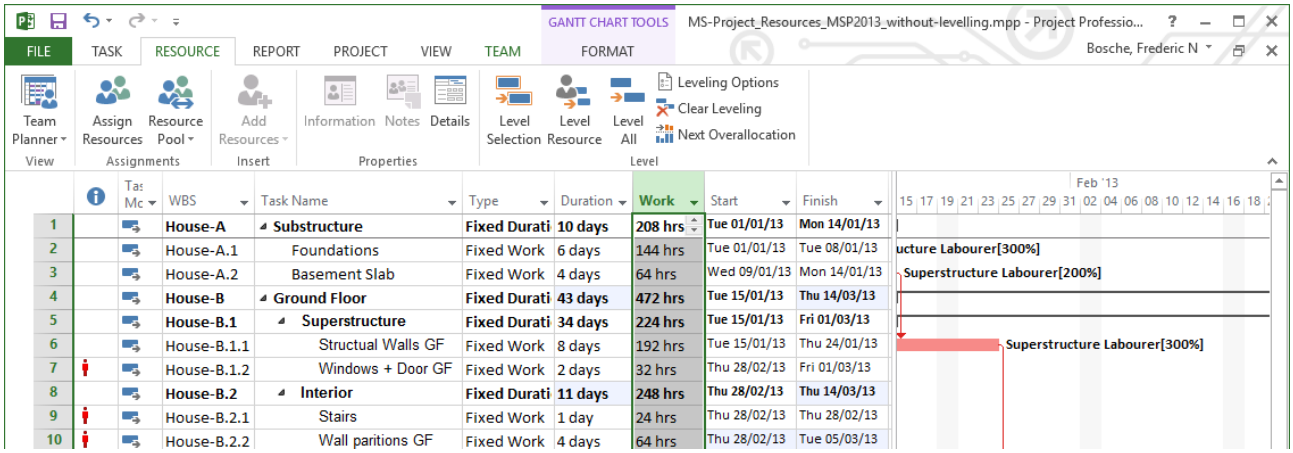


Figure 11: *Gantt Chart* view with the column *Work* added to the *Entry* table.

8.2.2 Fixed Duration, Fixed Work, Fixed Units, and Effort-driven Scheduling

Given Equation (1), MS Project enables a task/activity to be treated as having *Fixed Work*, *Fixed Duration* or *Fixed Units*. This *Task Type* can be defined for each individual task under the *Advanced* tab of the ***Task Information*** window. The default *Task Type* for all activities is defined in MS Project Options as explained in Section 2.

These three possible *Task Types* mean the following:

- *Fixed Work*: In this case, the amount of work required to complete this activity is considered fixed. This implies that, if you change the number of *Units* of the resource allocated to such an activity, then the *Duration* of the task will be automatically updated. And vice versa if you change the *Duration*.

  This option is preferred for "*effort-driven programming*" where an activity duration can be altered by reducing or increasing the number of resources assigned to it, or a target duration leads to a necessary amount of resources.

- *Fixed Duration*: In this case, the task duration is forced to be fixed. This implies that, if you change the number of allocated resource units, then the amount of work produced to complete the task will be automatically updated. And vice versa if you change the amount of work required.

  This option is preferred for "*time-driven programming*" where the project shouldn't be delayed and so conducting more work in a given task can only be achieved by increasing the number of resource *Units* (actually, this could also be achieved by changing their calendar and have them work more than 8 hours a day, or work on the weekend).

- *Fixed Units*: In this case, the number of resource units that will be used for this task is forced to be fixed. This implies that, if you change the amount of work required to be conducted, then the activity Duration will be automatically updated. And vice versa if you change the task duration.

  This option is preferred for "*resource-driven programming*", where the manager has fixed resource availability and must use them to their maximum efficiency.

To have easy access to the *Task Type* of each task/activity, I recommend you add the column *Type* in the *Entry* table used in the *Gantt Chart* view.

IMPORTANT NOTE: A very important thing to remember is that when you first create a new task, MS Project only applies the formula in Equation (1) once at least three of its parameters are available. Only then, does it take the *Task Type* into account for when you make a change to any of the values in those parameters. This means that you must ensure you enter the exact parameter values (*Duration*, *Units*, *Work*) the first time. Otherwise, if you come back later to correct one of the values, the behaviour of MS Project may not align with your expectations. This can unfortunately lead to great confusion.

8.2.3 Assigning Resources

To assign resources to a given task, you should use only the following two options (there is another way directly in the table, but this is actually likely to lead you into the kind of trouble described above):

- Open the ***Task Information*** window, and go to the *Resource* tab (see Figure 12); or
- Go to ***Resources*** ribbon and click on *Assign Resources*. This opens the ***Assign Resource*** window shown in Figure 13).

The two give access to more or less the same table. In this table, rows are used for the assignment of resources from the list of available resources defined in the *Resource Sheet* (see Section 8). To assign a resource in the ***Task Information*** window, click on the first empty cell in the *Resource Name* column and select in the dropdown menu the resource to be assigned. Then define the number of *Units* of that resource to be assigned to that task. In the ***Assign Resource*** window, the resources are normally already listed, one just needs to set the number of *Units* of each resource to be assigned.

In the ***Assign Resource*** window, it is also possible to define whether the resource is on *Request* or on *Demand (R/D)*. This is in relation to the resource substitution type. The *Request* setting indicates that the specified resource should be used unless the resource is overallocated. The *Demand* setting indicates that the specified resource must be used and not substituted. The blank setting (the default), indicates that any qualified resource can be assigned to the task. Don't worry about this option, it is very unlikely you'll ever have to use it.

Remember what is said in the IMPORTANT NOTE above. If you have entered Durations first, then the initial amount of resource *Units* that you will enter here will be used to calculate the amount of Work necessary for that task. If the task's *Task Type* is *Fixed Work*, then any subsequent change of the amount of *Units* will lead to a change in the task *Duration* that you had initially defined.

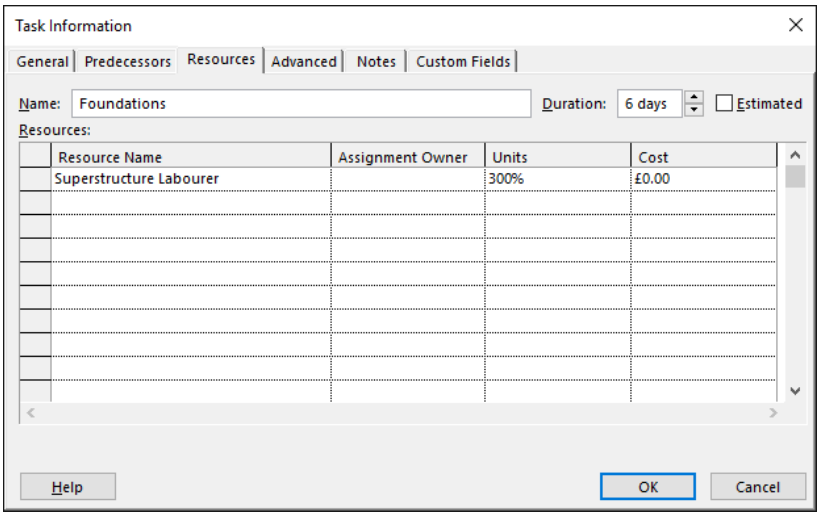


Figure 12: Task Information window – Resources tab.

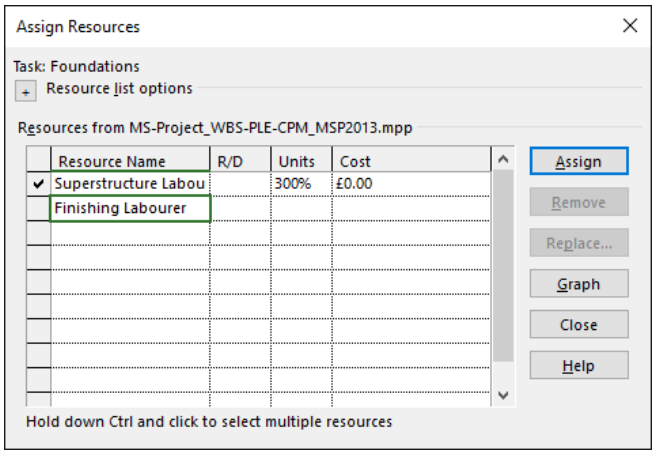


Figure 13: Assign Resources window.

Apply this:

Using the project information in the Support Project sheet (from Lab 1), do the following:

1. Go to the *Gantt Chart* view, select the *Entry* table and add the *Work* column next to the *Duration* column. The amount of Work should be zero because MS Project does not yet have enough information to calculate it automatically, or it has not been entered by the user.
2. Enter the resource requirements for each of the programme activities. Try using the two different windows as described above.
3. Once completed, change the *Task Type* of one task to *Fixed Work* and double the number of resource *Units* assigned to it; see how the *Duration* changes accordingly (while the amount of *Work* remains indeed unchanged). Undo this and alternatively change the *Task Type* of the task to *Fixed Duration* and double the number of resource *Units* assigned to it; see how this time, the amount of *Work* changes, while the *Duration* remains unchanged.
4. Undo what you did in the step 3 above before continuing.

IMPORTANT NOTE: As mentioned earlier, the *Task Type* selected enforces the default behaviour of MS Project for managing the task. Equation (1) involves three parameters (because MS Project considers the *Maximum Daily Workload* to be fixed). This means that, when the user later changes one of the two parameters that are not *Fixed*, then MS Project can automatically calculate the second non-fixed parameter. However, MS Project cannot know what the user wishes to do if s/he changes the parameter that is supposed to be fixed. For example, in the case of a task with *Fixed Duration*, if the user changes the *Duration*, should MS Project update the amount of *Work* or the amount of resource *Units*? Since MS Project is enable to make the decision, it will pick one but issue a warning to the user to enable them to exactly define which behaviour they expect. Figure 14 below shows one such warning.

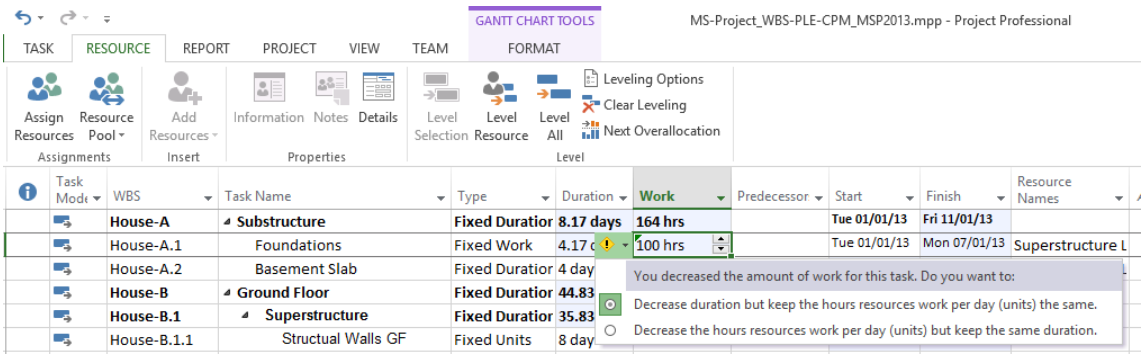


Figure 14: MS Project issues a warning after the user decreased the amount of work for a Fixed Work activity. In this case, the user is asked whether s/he wishes to see the *Duration* decreased or decrease the number of *Units*. By default, MS Project pixked the first option, but the user should (must) confirm this.

Apply this:

1. Take a task in your programme and set it to *Fixed Work*, and then change the number of *Work* for it. See the issued warning asking you to decide the behaviour to follow.
2. Undo what you did in the step 1 above before continuing.
3. Save your MS Project file with “before-levelling” added to its name.
4. I recommend you now set the *Task Type* for all tasks to *Fixed Work*.

9 Resolving Over-allocations and Levelling

One of the purposes of assigning resources to tasks is to analyse how the different resources are used over the duration of a project, and whether some of these resources are *over-allocated (*or *under-allocated).* In close relation to this is the issue of ensuring that resource demand is *levelled*. Over-allocation and Under-allocation are derived from the difference between resource availability and resource requirements.

9.1 Visualising Resource Demand over Time

In order to analyse the allocation of resources, open the *Resource Graph* view (for example by selecting that view in the ***View*** ribbon). The left part of the view enables you to change the resource under analysis (by scrolling over this area), and the right part shows the allocation of that resource. Note that the Timescale of these diagrams is never very good. I recommend you adjust it, e.g. with Months in the medium-tier and Days in the lower-tier (as shown in Figure 15).

Another interesting view is the *Resource Usage* view (selectable in the ***View*** ribbon) that summarizes the detailed usage of each resource per task. In this form and all other resource views, all overallocated and underallocated resources appear in red.

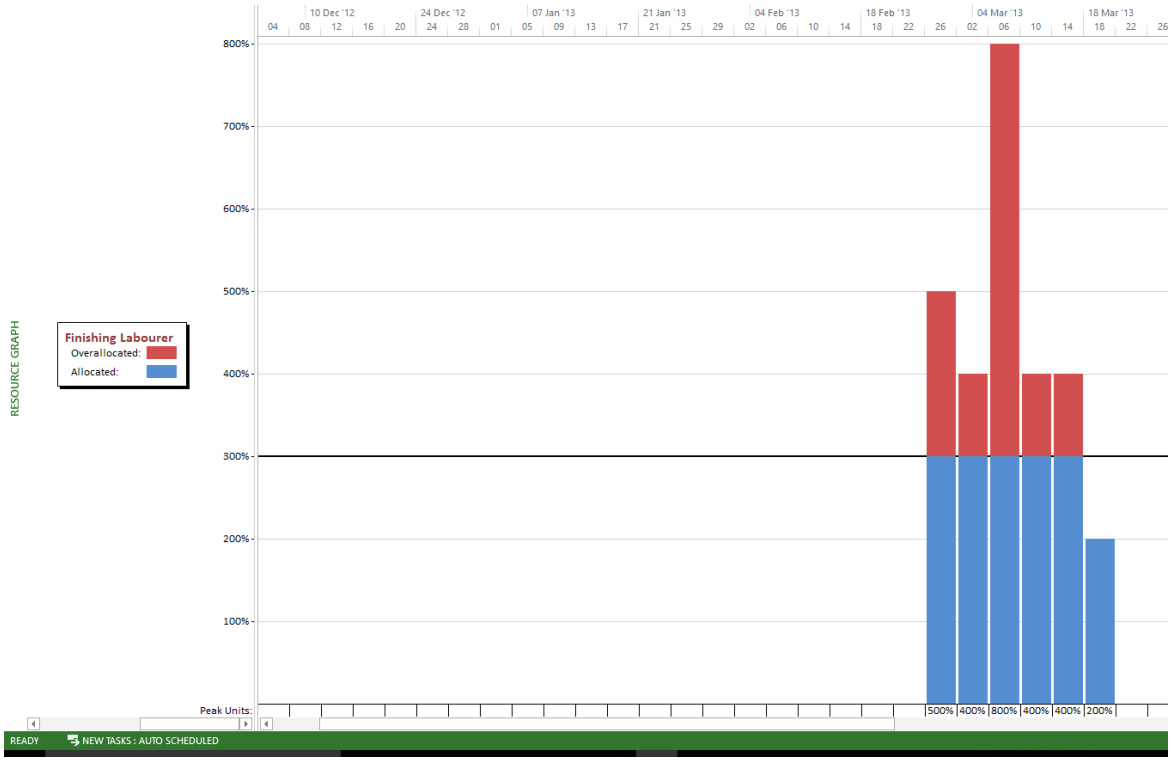


Figure 15: Resource Graph views for two.

Apply this:

Look at the resource graphs for the resources involved in your project. To better assess any overallocation, change the timescale to Month and Days.

9.2 Resolving Over-allocations and Levelling[2]

MS Project gives you several options to address overallocation and over-fluctuating resource demand. The first one below is cost-free, while the following two are not and so their selection should be made based on a cost-benefit analysis.

1. Level within slack: the preferred approach (because it is cost-free) to address overallocation and fluctuating resource demand is to *level* the demand for resources using the slack available in the programme.

   MS Project has a levelling functionality that is available under the ***Resource*** ribbon. To level within slack, click on *Leveling Options* to open the ***Resource Leveling*** window shown in Figure 16. Levelling can be manual or automated. Under Leveling Calculations, *Manual leveling* (the default) occurs only when you click on *Level Now*. Automatic levelling occurs instantaneously whenever you change a task or resource. Use *Automatic Leveling* if you want to reschedule tasks whenever resources are assigned more work than they have the capacity to complete – in that case, it is recommended to also uncheck the *Clear Leveling values before leveling* checkbox. I would however recommend never to use *Automatic Levelling* and always carefully resolve levelling manually when there is a need for it.

   Under *Leveling range*, select to level the entire project or to level only those tasks falling within a specific time range.

   Under the *Resolve Overallocations* box, several important options are available. In the *Leveling order* box, select:

   - *ID Only:* to level tasks in ascending order of their ID numbers before considering any other criteria;
   - *Standard:* to first examine predecessor dependencies, slack, dates, priorities, and constraints to discover whether and how tasks should be levelled (this is the default and preferred setting); or
   - *Priority, Standard:* to check task priorities first and only then examine the standard criteria.

   To prevent the finish date of your project from being delayed (i.e. not delay the critical activities), check the *Level only within available slack* checkbox.

   If you want to clear the previous levelling results before levelling again, then check the *Clear Leveling* checkbox.

   For levelling within slack, this is all you need to do and you may now click *Level All*. You should now see that the schedule has been updated and non-critical activities have been moved (some are now critical) to level resource demand, and in particular resolve overallocations as much as possible. Note

[2] In British English, "levelling" takes two "l", while in American English it takes only one. Since MS Project is an American product, I use the American spelling to fully match their writing in the software package.

that you may get error messages indicating that MS Project cannot level the entire schedule. This is because there is not enough slack in a schedule to resolve all overallocations.

2. Level by delaying project completion: In this case, it is allowed to alter critical activities. This will however lead to a delay in the project, which could have cost implications. Therefore, investigating this approach should be done within a cost-benefit analysis. This is done by following the same procedure as above, but unchecking the *Level only within available slack* checkbox. The following other options can then be considered.

   To allow levelling to adjust when a resource works on a task independently of other resources that are working on the same task, select the *Leveling can adjust individual assignments on a task* checkbox.

   If you want levelling to interrupt tasks by creating splits in the remaining work on tasks or resource assignments, select the *Leveling can create splits in remaining work* checkbox. If a resource is assigned to tasks concurrently beyond what the resource's schedule can handle, then a task that has remaining work can be split and worked on when the resource's schedule will allow it.

   To include *proposed* resources, select the *Level tasks with the proposed booking type* checkbox.

   Like before, to start the levelling you would now click on *Level Now*.

3. Set overtime: allow certain resources to work over-time during certain periods and can be used to address overallocation without delaying the completion of the project. Adding over-time essentially leads to an increase in the *maximum daily workload*, which enables a reduction in the required *Units* of that resource (see Equation (1)).

   Overtime can be assigned manually to a resource per activity. For this, go to the ***Resource Usage*** view, and change the table view to *Work*. You can change the amount of overtime allowed for this resource for this specific activity. Adding overtime should enable you to either reduce the number of units without changing the duration, or reduce the duration without changing the number of units. The former case directly help in addressing the overallocation. The latter case leads to increase slack in the programme that can then be used to address overallocation. The default behaviour followed by MS Project will depend on the task's *Task Type*.

   Alternatively, one could also change the working calendar of the resource.

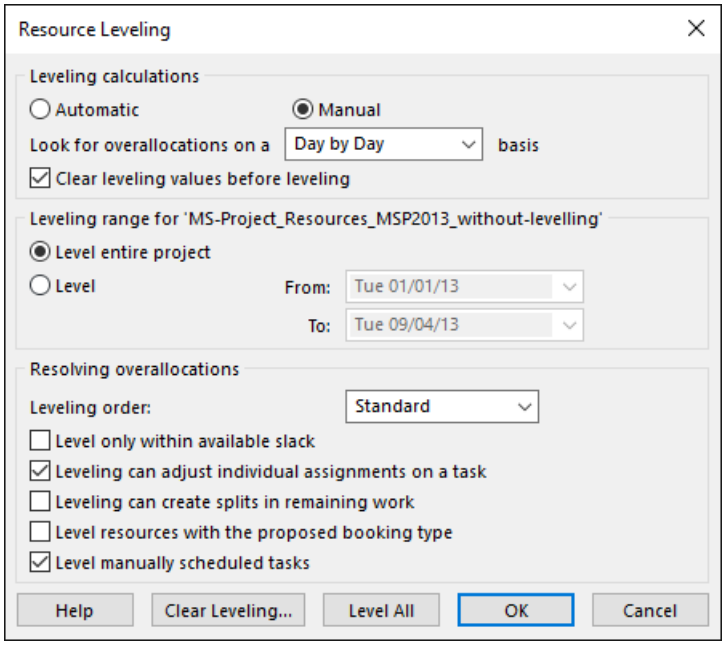
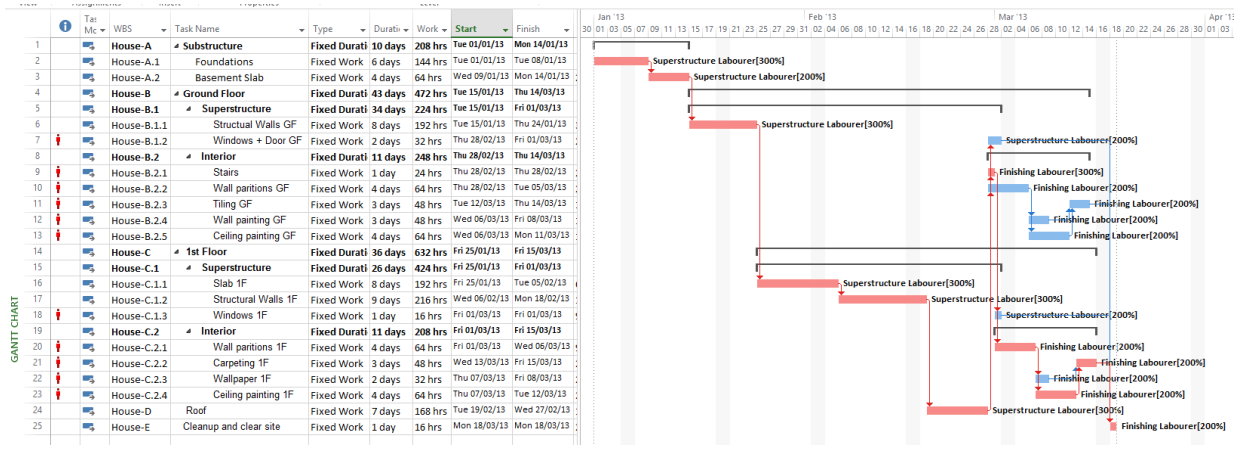


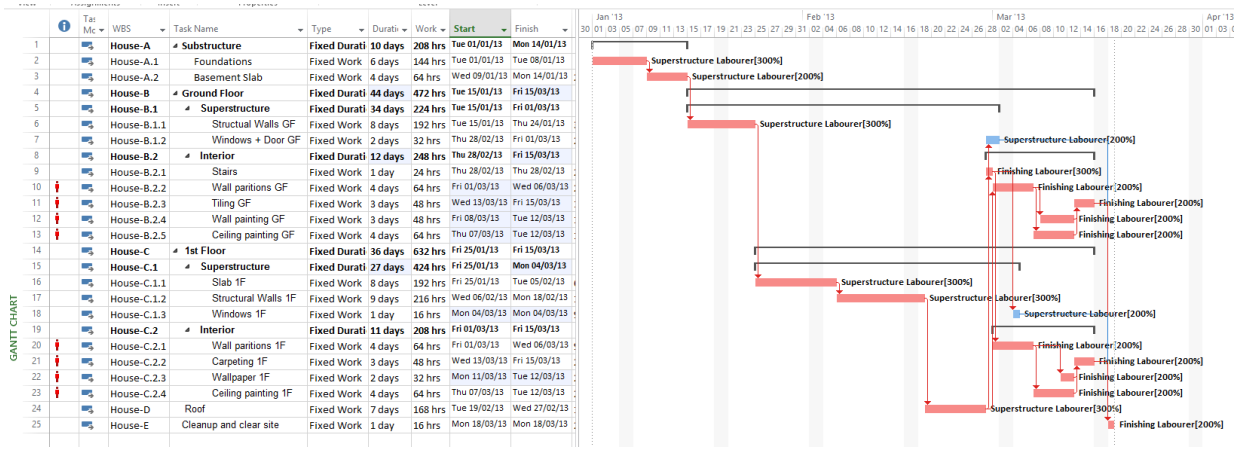
Figure 16: Resource Leveling window.

Apply this:

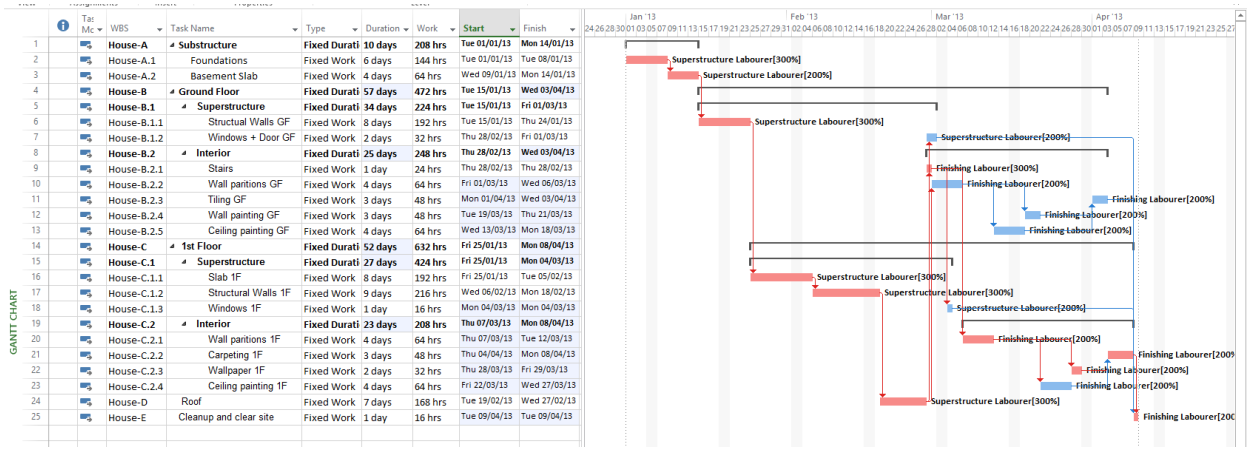
1. Level your programme within slack. See how the schedule is changed and look at how the resource demand profiles also changed. The results should correspond to those in Figure 17(b), Figure 18(b), Figure 19(b).
2. Level your programme to fully resolve all overallocations by altering also critical activities (don't use overtime). Use the same settings as in Figure 16. Again, see how the schedule is changed and look at how the resource demand profiles also changed. The results should correspond to those in Figure 17(c), Figure 18(c), Figure 19(c).



(a)



(b)



(c)

Figure 17: Programme before levelling (a), after levelling within slack (b), and finally after levelling without slack constraint (c).

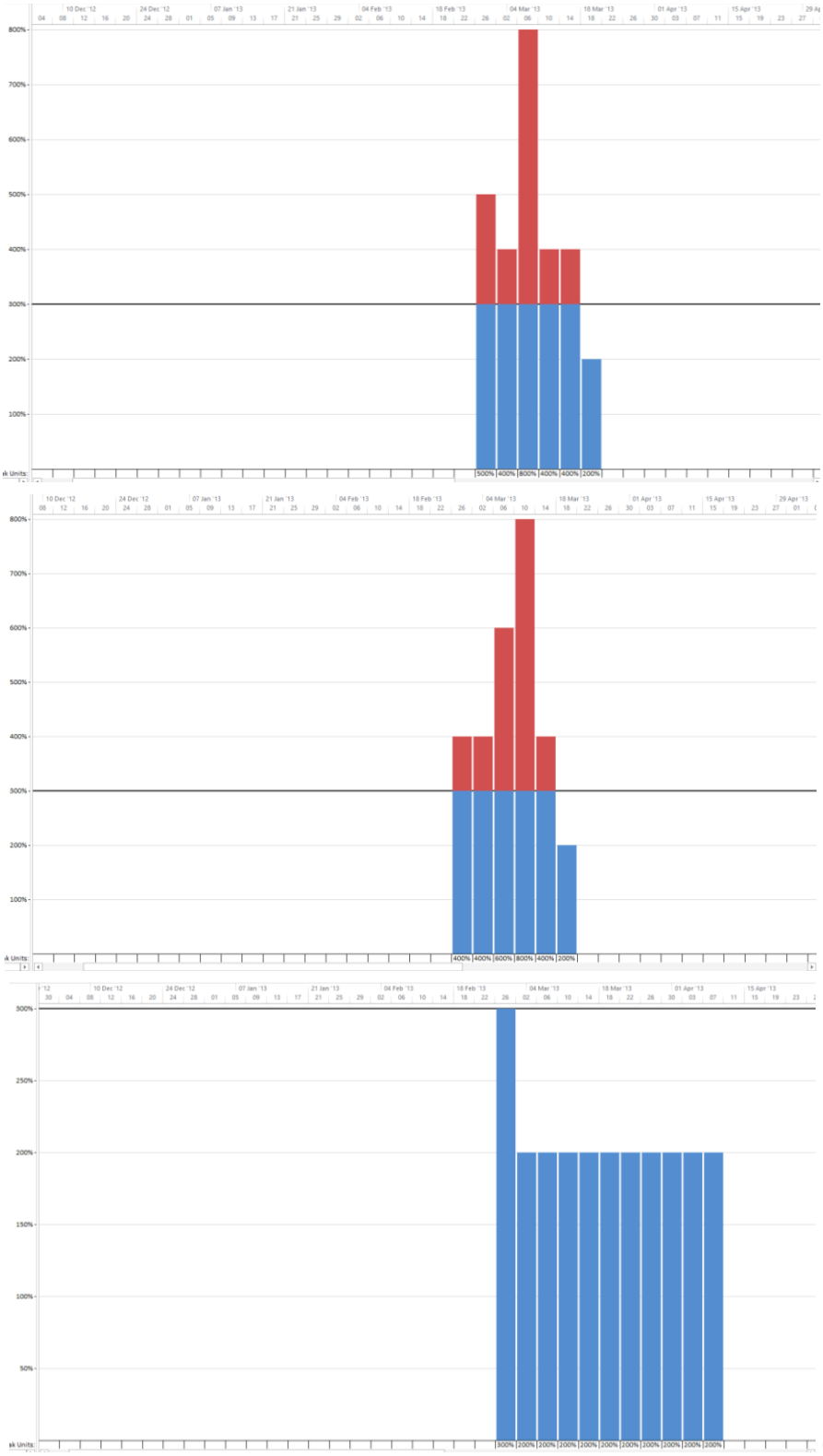


Figure 18: Resource Graph before levelling (a), after levelling within slack (b), and finally after levelling without slack constraint (c).

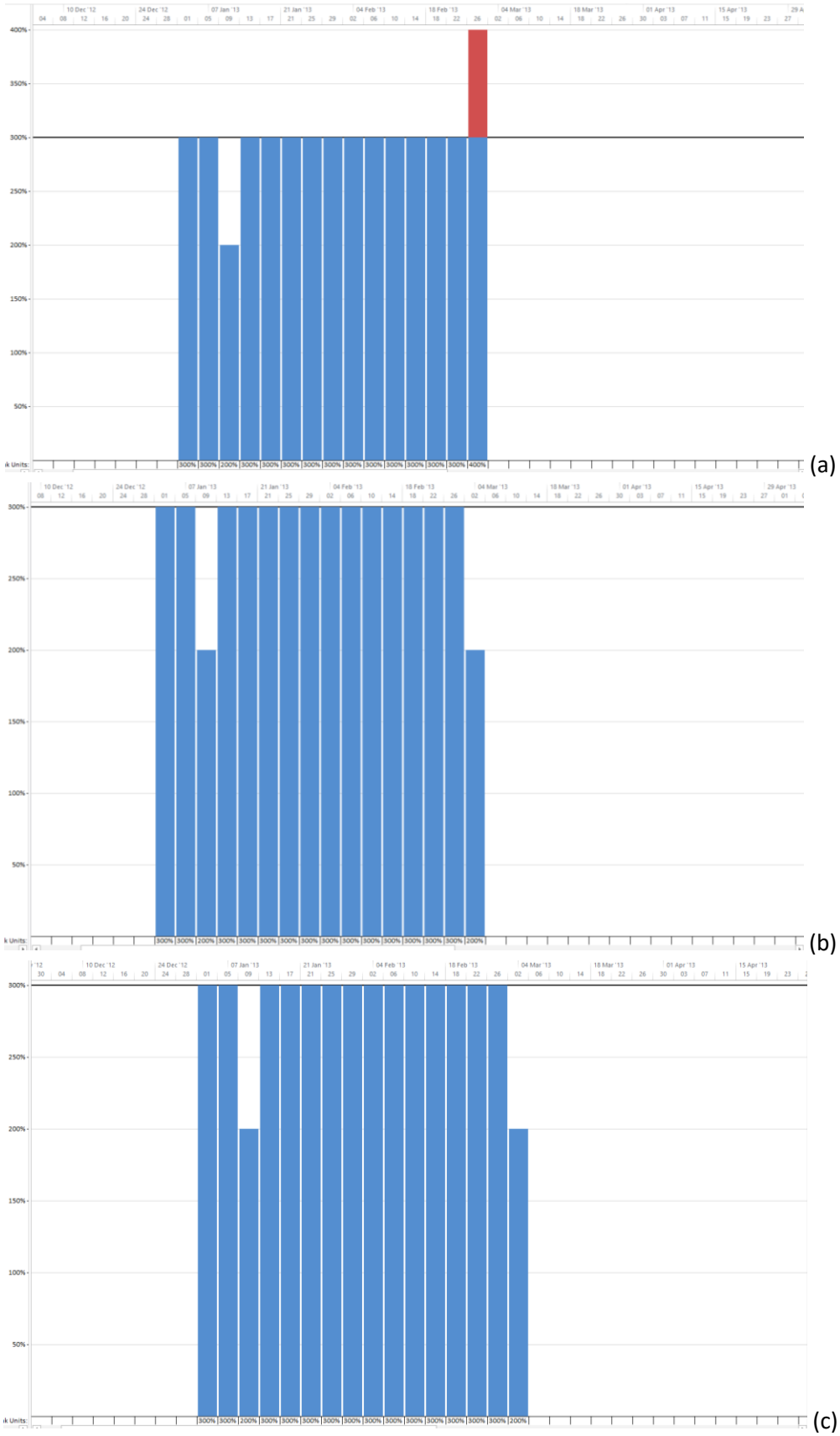


Figure 19: Resource Graph before levelling (a), after levelling within slack (b), and finally after levelling without slack constraint (c).

10 Entering Cost Information

MS Project enables the user to enter two types of costs:

- *Resource Costs*: that are associated to the usage of each resource.
- *Other Costs*: that may not be easily attributable to any resource

The following two sub-sections detail how these costs are entered.

10.1 Resource Costs

To enter resource cost information, open the ***Resource Sheet*** view (see Figure 20). The table contains four columns that relate to costs:

- *Std. rate*: Standard Rate is the "regular time" rate for a resource. It can be per hour, week, month, or year. For most projects, this is the prime driver of project cost calculations.
- *Ovt. Rate*: Overtime Rate is the overtime rate for this resource. Remember that MS Project does not know when overtime starts. It calculates overtime cost from values you enter in the *Overtime* field.
- *Cost/Use*: Per Use Cost defines a fixed cost associated with the use of a resource (fixed cost independent of the duration of use). This might be a contract initiation charge, or you might use it for resources that work on a flat fee or per-job basis instead of by the hour. BE CAREFUL, this value gets added to the Cost field of every task to which the resource is assigned. The *Use* in this case is not use on the project, but on each task. So, if a resource with a Per Use Cost value of £100 is assigned to three tasks, the project will see a cost related to Per Use Cost of this resource in the amount of £300.
- *Accrue At*: The Cost Accrual list box lets you define how costs for that resource should be accrued against a task. The choices for this field are: *Start*, *Prorated*, and *End*. For example, you might use *Prorated* for an in-house resource that is paid by the hour. Or you could use *End* for a resource that is not paid until the resource's work is completed. This field helps you control how the program calculates cash flow for the project (see Section 12.1). The most common setting here is *Prorated*.

Basic cost calculations for a task occur at the assignment level where:

$$Cost = (Regular\ Work * Standard\ Rate) + (Overtime\ Work * Overtime\ Rate) + Cost\ Per\ Use \quad (1)$$

10.2 Other Costs

There are costs that may not be easily attributable to any resource specifically identified in the programme. These additional costs can nonetheless be entered as follows.

Go to the ***Gantt Chart*** view, and change the table format to the *Cost*. While the table is discussed in more detail in Section 10.3 below, it contains two columns related to Fixed Costs:

- *Fixed Cost*: Fixed non-resource (non-labour) cost associated to the task.
- *Fixed Cost Accrual*: The method defining how fixed costs should be accrued for the task. The choices are the same as those for *Resource Costs* (as seen in Section 10.1): *Start*, *Prorated*, and *End*.

Apply this:

Using the project information in the Support Project sheet (from Lab 1.1), do the following:

5. Enter all resource costs.
6. Enter all non-resource costs.

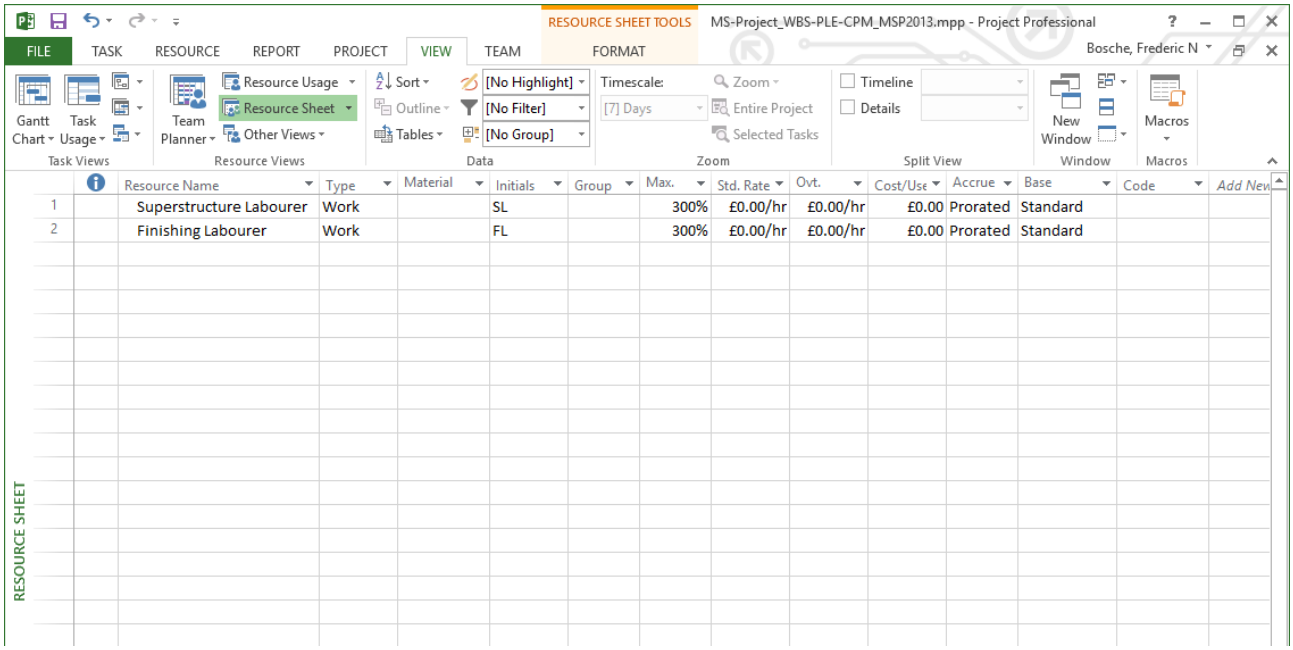


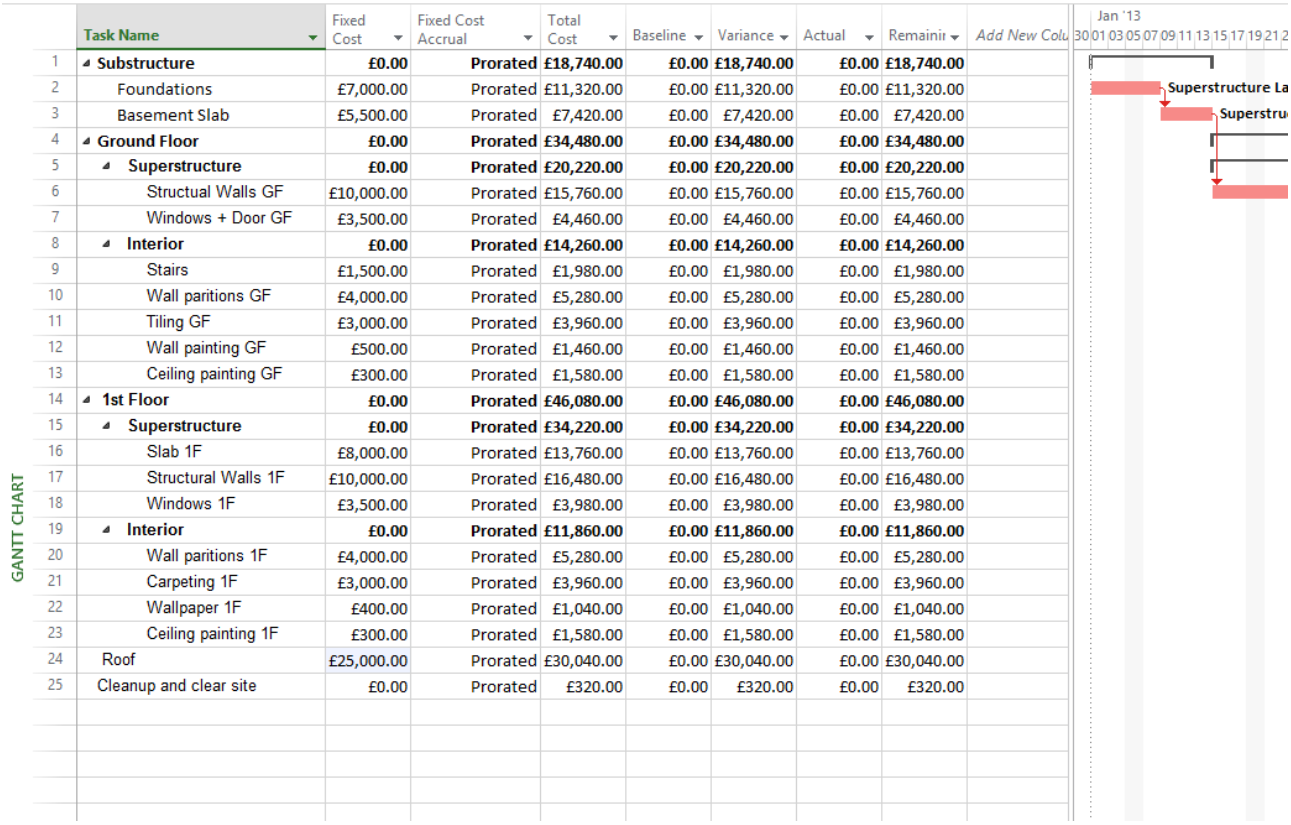
Figure 20: Resource Sheet view.

10.3 Budget

Open the ***Gantt Chart*** view with the *Cost* table (as before). The table shows you the accumulated cost for all tasks, but also at all levels of the WBS.

The table has seven cost columns (see Figure 21), six of which provide summarized cost information:

- *Fixed Cost*: Fixed non-resource cost associated to the task.
- *Fixed Cost Accrual*: The method defining how fixed costs for should be accrued for the task. The choices are the same as those for *Resource Costs* (as seen in Section 10.1): *Start*, *Prorated*, and *End*.
- *Total Cost*: The total cost of the task, including resource costs and fixed costs.
- *Baseline*: see **Lab 2**.
- *Variance*: see **Lab 2**.
- *Actual*: see **Lab 2**.
- *Remaining*: see **Lab 2**.



	Task Name	Fixed Cost	Fixed Cost Accrual	Total Cost	Baseline	Variance	Actual	Remainir	Add New Colu
1	**Substructure**	**£0.00**	**Prorated**	**£18,740.00**	**£0.00**	**£18,740.00**	**£0.00**	**£18,740.00**	
2	Foundations	£7,000.00	Prorated	£11,320.00	£0.00	£11,320.00	£0.00	£11,320.00	
3	Basement Slab	£5,500.00	Prorated	£7,420.00	£0.00	£7,420.00	£0.00	£7,420.00	
4	**Ground Floor**	**£0.00**	**Prorated**	**£34,480.00**	**£0.00**	**£34,480.00**	**£0.00**	**£34,480.00**	
5	**Superstructure**	**£0.00**	**Prorated**	**£20,220.00**	**£0.00**	**£20,220.00**	**£0.00**	**£20,220.00**	
6	Structual Walls GF	£10,000.00	Prorated	£15,760.00	£0.00	£15,760.00	£0.00	£15,760.00	
7	Windows + Door GF	£3,500.00	Prorated	£4,460.00	£0.00	£4,460.00	£0.00	£4,460.00	
8	**Interior**	**£0.00**	**Prorated**	**£14,260.00**	**£0.00**	**£14,260.00**	**£0.00**	**£14,260.00**	
9	Stairs	£1,500.00	Prorated	£1,980.00	£0.00	£1,980.00	£0.00	£1,980.00	
10	Wall paritions GF	£4,000.00	Prorated	£5,280.00	£0.00	£5,280.00	£0.00	£5,280.00	
11	Tiling GF	£3,000.00	Prorated	£3,960.00	£0.00	£3,960.00	£0.00	£3,960.00	
12	Wall painting GF	£500.00	Prorated	£1,460.00	£0.00	£1,460.00	£0.00	£1,460.00	
13	Ceiling painting GF	£300.00	Prorated	£1,580.00	£0.00	£1,580.00	£0.00	£1,580.00	
14	**1st Floor**	**£0.00**	**Prorated**	**£46,080.00**	**£0.00**	**£46,080.00**	**£0.00**	**£46,080.00**	
15	**Superstructure**	**£0.00**	**Prorated**	**£34,220.00**	**£0.00**	**£34,220.00**	**£0.00**	**£34,220.00**	
16	Slab 1F	£8,000.00	Prorated	£13,760.00	£0.00	£13,760.00	£0.00	£13,760.00	
17	Structural Walls 1F	£10,000.00	Prorated	£16,480.00	£0.00	£16,480.00	£0.00	£16,480.00	
18	Windows 1F	£3,500.00	Prorated	£3,980.00	£0.00	£3,980.00	£0.00	£3,980.00	
19	**Interior**	**£0.00**	**Prorated**	**£11,860.00**	**£0.00**	**£11,860.00**	**£0.00**	**£11,860.00**	
20	Wall paritions 1F	£4,000.00	Prorated	£5,280.00	£0.00	£5,280.00	£0.00	£5,280.00	
21	Carpeting 1F	£3,000.00	Prorated	£3,960.00	£0.00	£3,960.00	£0.00	£3,960.00	
22	Wallpaper 1F	£400.00	Prorated	£1,040.00	£0.00	£1,040.00	£0.00	£1,040.00	
23	Ceiling painting 1F	£300.00	Prorated	£1,580.00	£0.00	£1,580.00	£0.00	£1,580.00	
24	Roof	£25,000.00	Prorated	£30,040.00	£0.00	£30,040.00	£0.00	£30,040.00	
25	Cleanup and clear site	£0.00	Prorated	£320.00	£0.00	£320.00	£0.00	£320.00	

Figure 21: *Gantt Chart* view with the *Cost* table.

Apply this:

Explore the Cost table and identify the Level 1 Work Package with the highest costs.

11 Programme Visualisation and Printing

The visualization (and printing) of a MS Project programme is not always a straight-forward matter. In MS Project, when you print or print preview a Gantt Chart view or a Usage view (see Section 9.1), task bars or columns of data may be printed over multiple pages, which makes the printout very hard to read.

In this section, you will learn how to customise the programme *Timescale* (used to display the Gantt Chart programme) and how it can be altered to optimally fit the screen or the printed page. Then, the *Timeline* is presented that can be used to present a summary of a project in visually pleasing manner. This can be useful in case the audience is not familiar with Gantt Charts, and to provide just a simple summary of the programme.

11.1 Timescale

Figure 22 shows a timescale as it may appear on screen on top of the Gantt Chart diagram in the Gantt Chart view. Its format may not really suit your needs, and tends to be set somewhat randomly. One can zoom in and out of the timescale by holding the Control button on the keyboard and using mouse scroll (while the cursor is above the Gantt Chart diagram). However, this does not provide a very precise means to achieve adequate levels of zooms. The zoom "+" and "-" buttons at the bottom-right of the MS Project window, only provide the same inadequate level of precision. It is nonetheless possible to both achieve adequate levels of zoom and have a meaningful (i.e. easily understandable) Timescale above the Gantt Chart diagram.

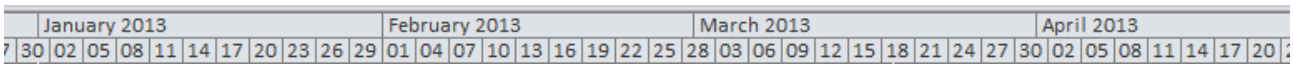


Figure 22: An example timescale as appearing above a Gantt chart.

By right-clicking on the timescale and select *Timescale...* you open the ***Timescale*** setup window shown in Figure 23. If you select the *Top Tier* tab, it contains two sections: *Bottom tier formatting* and *Timescale options*. If you then click on the *Middle Tier* and *Bottom Tier* tabs, you will see that the *Timescale options* section remains and only the first section changes.

The *Timescale options* section is used to set the general format of the timescale. The dropdown box *Show* enables the user to select whether the timescale should show one, two or three tiers. For example, the timescale shown in Figure 22 includes only the medium and bottom tiers (default setting). The dropdown box Size then enables you to do an overall horizontal scaling the timescale.

Now, for each of the first three tabs, the options in the *[Top/Medium/Bottom] tier formatting* sections can be set, if the tier is included in the selection made in the *Show* dropdown box. For example, Figure 23 shows the setting of the Medium Tier of our timescale. In this view, the dropdown box *Units* is used to set the scale (units) of time to be displayed at that tier. Below it, *Count* enables you to set whether you want to see each increment of that scale of time or skip some. For example, with the medium tier set to "month", a *Count* value of "1" will show every month on the scale (as shown in Figure 22); setting the Count to "3" would show only one in three months (i.e. January, April, July, October). Finally, the *Label* dropdown box gives you a range of options on the format to use to display that time scale.

Once you are pleased with the timescale, it is likely that the project will still not fully fill the available space on screen. To address this, go back to the ***Timescale*** setup window and change the *Size* percentage in the *Timescale options* section. If the programme was too small, increase the percentage; otherwise decrease it.

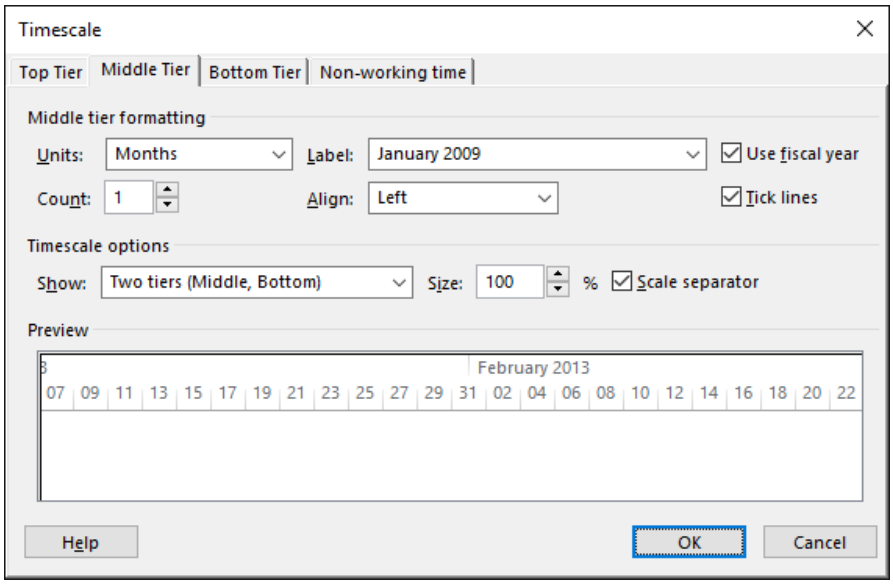


Figure 23: The Timescale setup window.

Apply this:

Set up a three-tier timescale showing the years, months, and weeks respectively. The year should appear in the number format "2001", the month in the format "Jan", and the week in the number format "1,2,3,...52". Once this is done, set the size percentage so that the programme nicely uses all the left-to-right space available on screen.

11.2 Printing

For improving the printing behaviour (e.g. having the whole project printed on one page), you can scale the entire project (not just the timescale portion as described in Section 11.1). To do this, follow these steps:

1. On the ***File*** ribbon, click *Print*, and then click on the link *Page Setup*.
2. Click the *Page* tab.
3. Choose the scaling option you want to use:
    a. Either scale the whole project as a percentage or define exactly the number of pages you wish the project to be printed onto.
4. Click *OK* and then *Print* your project.

11.3 Timeline

When you start MS Project, the *Timeline* is already visible between the ribbons and the main display area (Gantt Chart view), as shown in Figure 24. The Timeline provides an alternative means of visualisation the project. It can be hidden or shown again by (un)checking the *Timeline* checkbox in the ***View*** ribbon.

By clicking on the Timeline area, the **Timeline Format** ribbon appears (see Figure 24). To add tasks to the Timeline, select *Existing Tasks* in the ribbon, and the ***Add Tasks to Timeline*** window shown in Figure 25 appears. The window lists all the activities and Work Packages currently existing in the programme and each has a checkbox that can be checked to add it to the timeline. Figure 26 shows an example of Timeline after having added some existing tasks to it.

Once the activities have been added, the presentation of the timeline can be changed. For example, you can change the line on which each task appears by dragging the task to the line you wish to use it (dragging it below the diagram enables the creation of an additional line). The visual appearance of the timeline and individual tasks can also be changed by selecting them and using the relevant formatting options in the ribbon. Figure 26 shows an example of completed Timeline.

Apply this:

Using your project, create a timeline that contains all the Level 2 WPs (i.e. “substructure”, etc.). Change the date format to “01 Jan” and change the colour of the activities and background of the timeline.

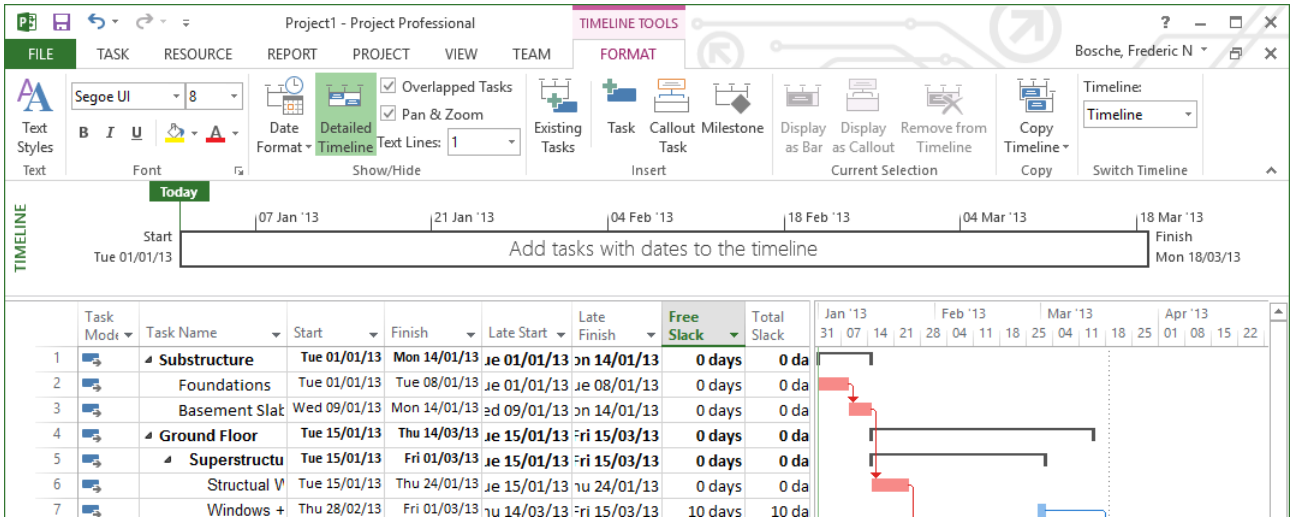


Figure 24: The (empty) Timeline between the ribbons and the main display area. The ribbon shown here in the Timeline Tools ribbon (that appears when clicking in the Timeline area).

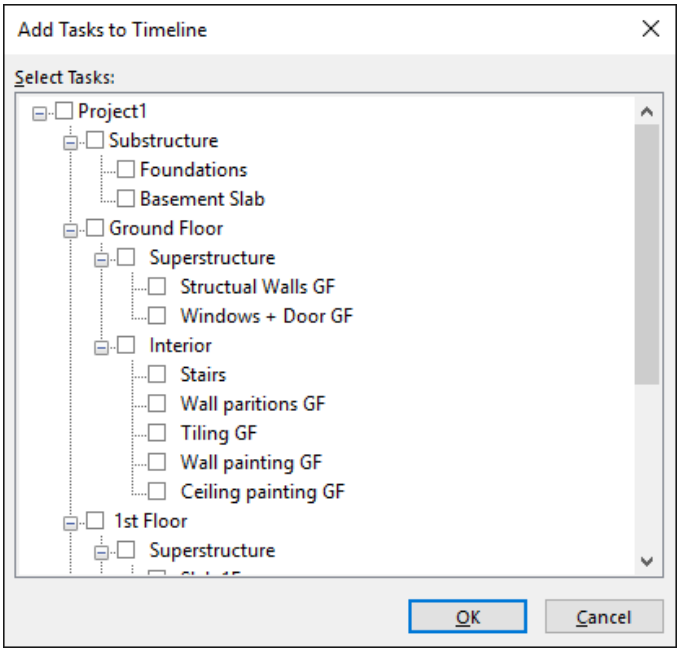


Figure 25: The window Add Tasks to the Timeline

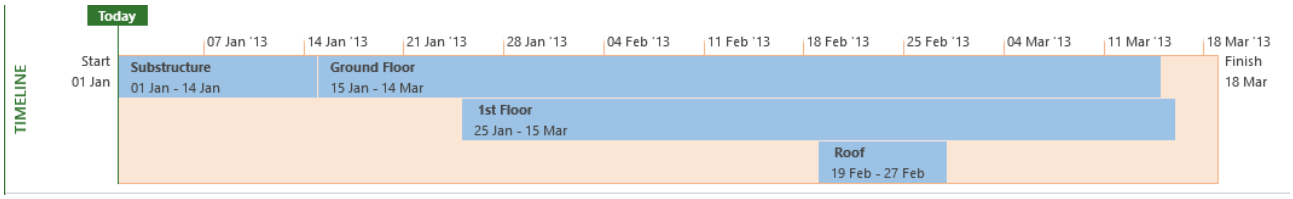


Figure 26: Example Timeline.

12 Producing Reports

MS Project offers better ways to save the programme information than taking screenshots. The ***Report*** ribbon contains different types of reports organised in two main types:

- *View Reports*: which are pre-defined standard reports that are viewed on the screen and can be printed. For example, Figure 27 shows the *Cost Overview* dashboard that is found under the *Dashboard* dropdown menu, and Figure 28 shows the *Task Cost Overview* report that is found under the *Costs* dropdown menu.
- *Export Reports (Visual Reports)*: these are graphical reports that are exported in the form of Excel or Visio files that can be customized to show only what is desired. These are really useful reporting options. For example, Figure 29 shows the customizable *Visual Report* called *Baseline Work Report* opened in Excel.

All reports can also be saved as PDF (or XPS) by going to **File**→Export and selecting *Create XPS/PDF*.

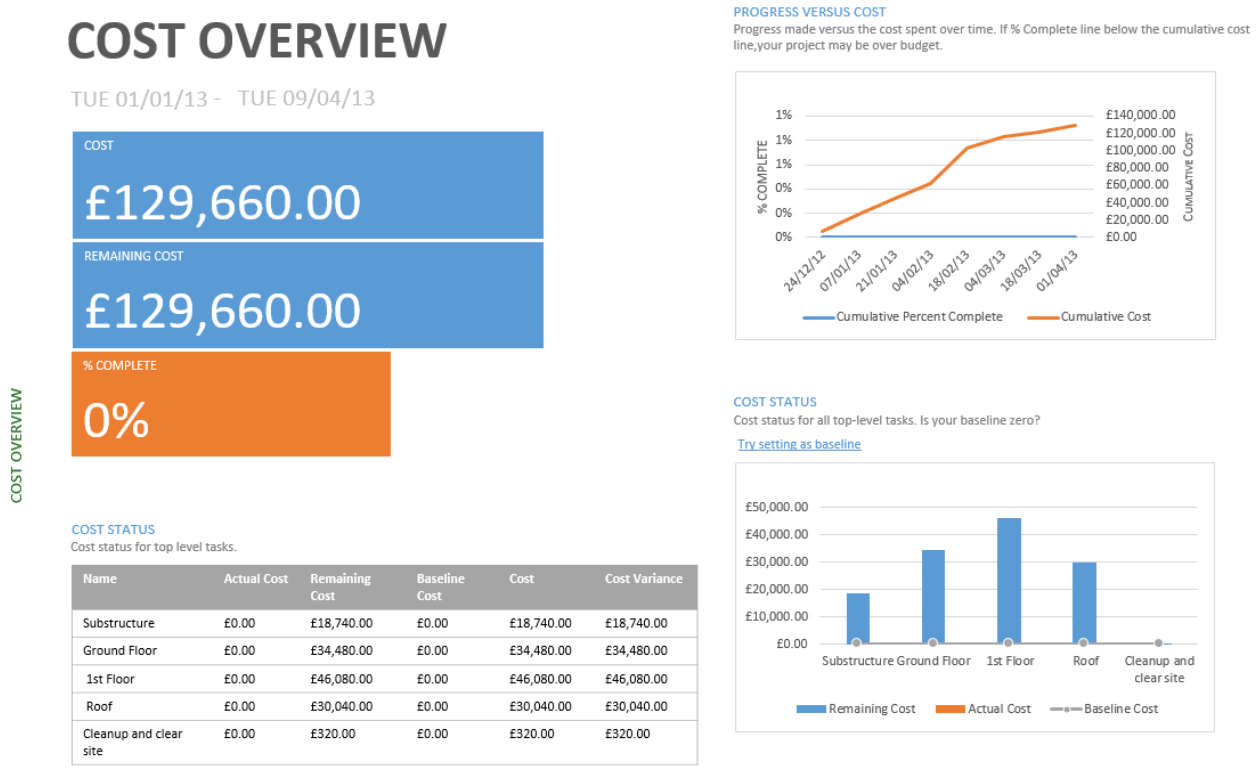


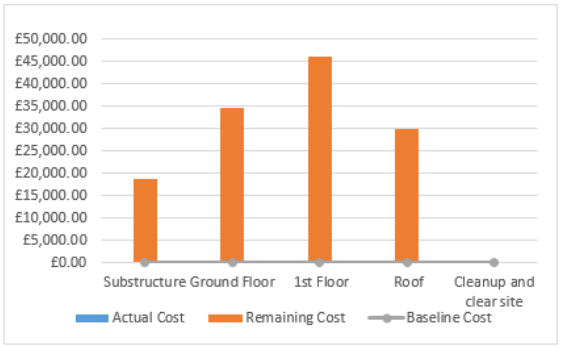
Figure 27: *Cost Overview* dashboard.

TASK COST OVERVIEW

TASK COST OVERVIEW

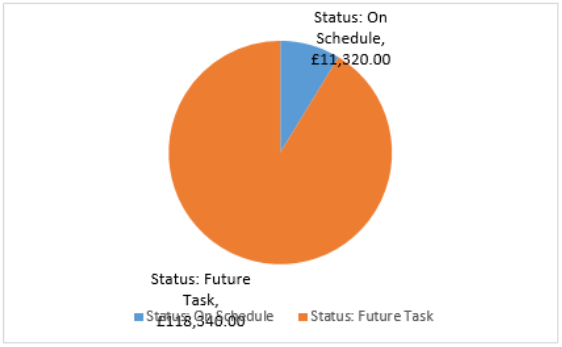
COST STATUS

Cost status for top-level tasks.



COST DISTRIBUTION

How costs are spread out amongst tasks based on their status.



COST DETAILS

Cost details for all top-level tasks.

Name	Fixed Cost	Actual Cost	Remaining Cost	Cost	Baseline Cost	Cost Variance
Substructure	£0.00	£0.00	£18,740.00	£18,740.00	£0.00	£18,740.00
Ground Floor	£0.00	£0.00	£34,480.00	£34,480.00	£0.00	£34,480.00
1st Floor	£0.00	£0.00	£46,080.00	£46,080.00	£0.00	£46,080.00
Roof	£25,000.00	£0.00	£30,040.00	£30,040.00	£0.00	£30,040.00
Cleanup and clear site	£0.00	£0.00	£320.00	£320.00	£0.00	£320.00

Figure 28: *Task Cost Overview* report.

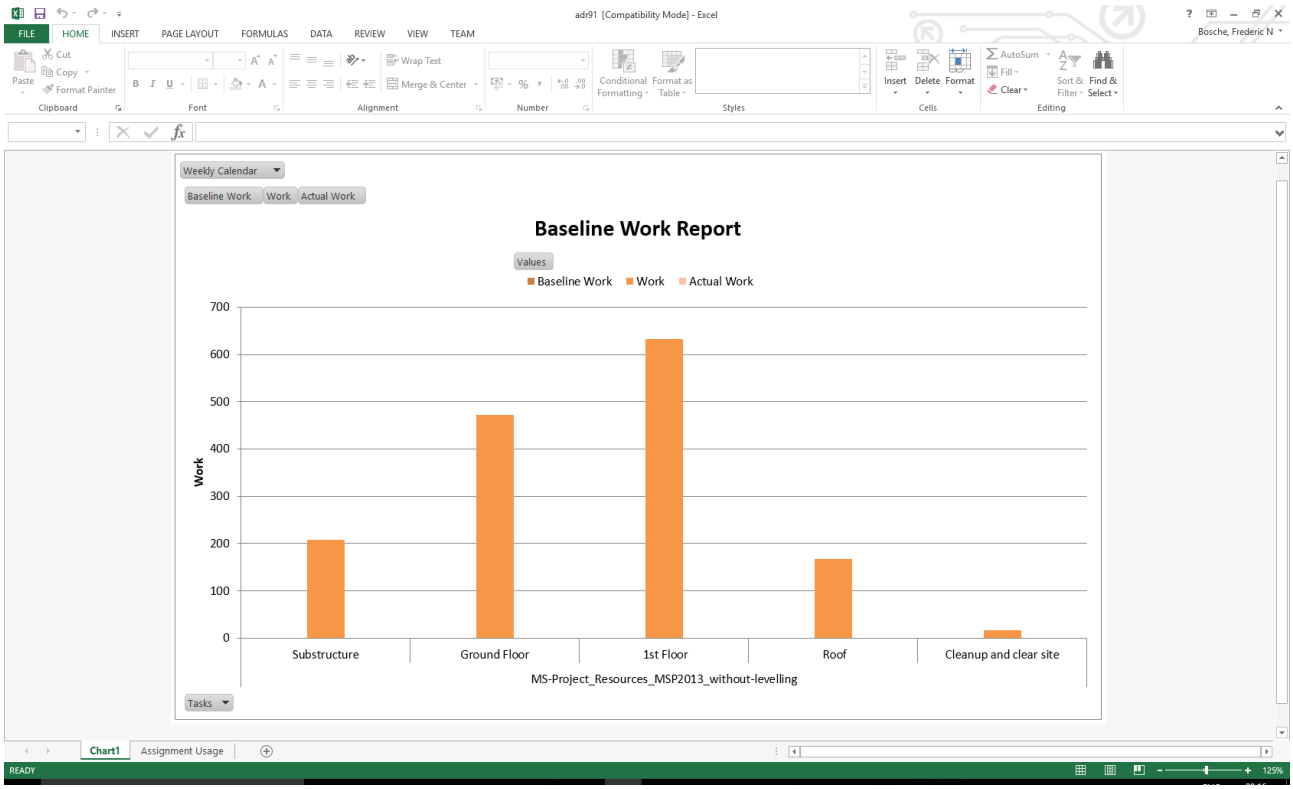


Figure 29: Baseline Work visual report.

12.1 Report Information Granularity

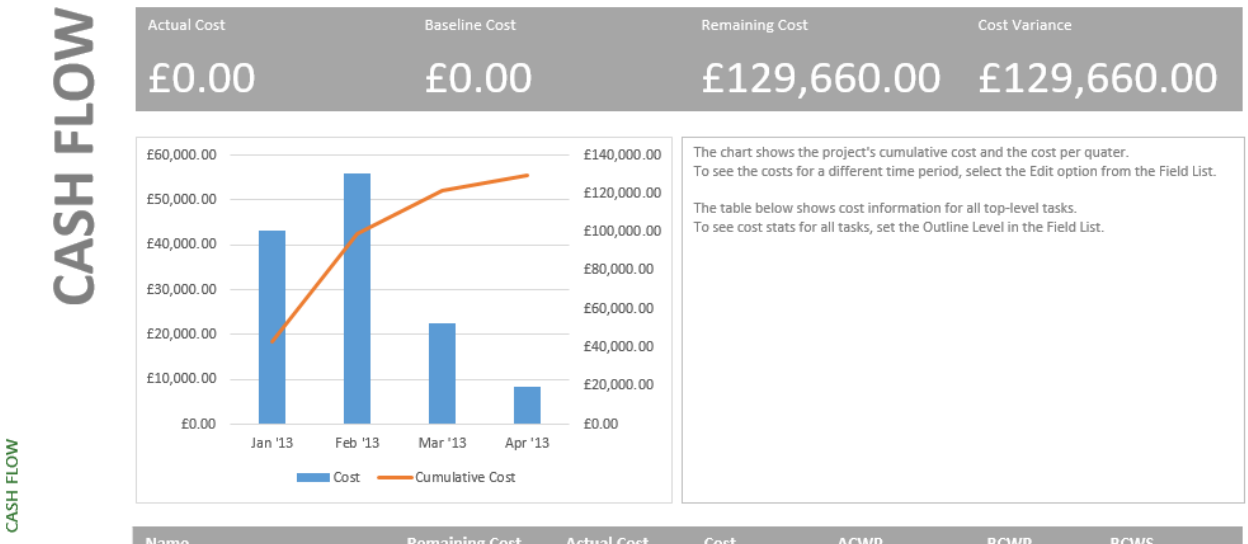
Some reports are also available that provide different summaries of information, such as the *Task Cost Overview* report (under the *Cost* dropdown menu) or the *Baseline Cost Report* visual report.

In reports like these, you can change the granularity in time or in the WBS in which the data should be shown.

For example, the upper-left graph in the *Task Cost Overview* (Figure 28) shows the cost per WP, by default at Level 1. But, you can change the level of granularity by selecting the graph and changing the *Outline Level* to Level 2 (or more). You can tick the box *Show Hierarchy* so that the horizontal axis shows the whole WBS hierarchy.

In another example, the upper-left graph in the *Cash Flow* report (Figure 30) shows data per quarter or month. To change this, select the graph, and in the menu appearing on the right, change Select Category to Time and click Edit. You can then change the units to whatever you need (Figure 30 shows in Months).

Note: MS Project does not provide means to reconciliate Cost and Value, as this is more an accounting matter than a project management one.



Name	Remaining Cost	Actual Cost	Cost	ACWP	BCWP	BCWS
Substructure	£18,740.00	£0.00	£18,740.00	£0.00	£0.00	£0.00
Ground Floor	£34,480.00	£0.00	£34,480.00	£0.00	£0.00	£0.00
1st Floor	£46,080.00	£0.00	£46,080.00	£0.00	£0.00	£0.00
Roof	£30,040.00	£0.00	£30,040.00	£0.00	£0.00	£0.00
Cleanup and clear site	£320.00	£0.00	£320.00	£0.00	£0.00	£0.00

Figure 30: *Cash Flow* report.

Apply this:

1. Generate a *Baseline Work* visual report.
2. Click on the *Assignment Usage* tab. There you see the *PivotTable* that is used to fill the diagram in the first tab. In the table, expand all Task 1 tasks to show all Task 2 ones. Then, return to the first tab and see how the diagram was updated accordingly to show the new level of detail.
3. The diagrams above add up all the work for all resources. To see it split per resource, return to the *Assignment Usage* tab and the *PivotTable Fields* area (right column) add the R*esources* to the *Rows* (by dragging them in that area). See what happens to the diagram in the first tab.
4. Return to the *Assignment Usage* tab, and now move the *Resources* from beneath *Tasks* to above *Tasks* within the *Rows* section. See again how the diagram in tab 1 updated accordingly.
5. Investigate the *Cost Overview* dashboard. What is the overall cost of the project? Now, change the breakdown of the Cost Status sub-table to show level 2 tasks (with hierarchy). Finally, change the time resolution of the Progress vs Cost diagram to show the information per week with a format (1, 2, … 52).
6. Investigate the *Task Cost Overview* and *Cash Flow* reports for your project. Save one of the two reports in PDF format.